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RESEARCH PAPER ON
SEA BELT & DIVERT SYSTEM
SYSTEM

Smart Wearable Safety System for Fishermen

PROJECT SUBMITTED BY

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “**SMARTSEA BELT: Design and Development of a Marine SafetyBelt Using NavIC GPS, LoRa Communication, Health Monitoring, and Automatic Inflatable FlotationSystem for Fishermen Safety**” is a bonafide work carried out by Fathima Rahna R, Celano S, InbhaBenisha J of the Department of Master of Business Administration, Rohini College of Engineering and Technology, during the academic year 2025–2027, under the guidance and supervision of the college.

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DECLARATION

We hereby declare that this dissertation entitled “SMARTSEA BELT: Design and Development of an Marine Safety Belt Using NavIC GPS, LoRa Communication, Health Monitoring, and Automatic Inflatable Flotation System for Fishermen Safety” is an original work carried out by us under the guidance of Dr. R. Benisha, Head of the Department of Biomedical Engineering, Rohini College of Engineering and Technology. This work has not been submitted either wholly or partially to any other university or institution for the award of any degree or diploma. The information presented in this dissertation has been collected from reliable sources, experimental work, and published research papers, and all such sources will be properly acknowledged in the final dissertation.

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ABSTRACT

The Smart Sea Belt is a proposed intelligent marine safety system designed to improve the safety of fishermen and other small-vessel occupants during emergency situations at sea. Marine environments can expose individuals to risks such as accidental falls overboard, loss of balance, delayed emergency communication, sudden health-related incidents and difficulty in reaching a safe location after an emergency.

The proposed system integrates wearable sensing, wireless communication, emergency flotation and an automatic rudder diversion mechanism into a coordinated safety architecture. Each Smart Watch unit can monitor parameters such as heart rate and motion and provides an SOS interface. Bluetooth Low Energy (BLE) is used for communication between wearable units and a centralized Boat Hub. The Boat Hub combines the wearable data with GPS information and forwards relevant emergency information through LoRa communication to a Base Unit. The Base Unit supports monitoring and emergency alerts.

For an emergency involving a person overboard or a critical safety event, the system is designed to support two physical safety responses: an automatically deployable flotation mechanism and an automatic rudder diversion mechanism. The flotation subsystem is intended to assist the person in maintaining buoyancy, while the rudder mechanism is intended to redirect the boat toward the relevant emergency location or safer recovery direction, subject to the vessel's operating conditions and control logic.

The proposed architecture follows the sequence “Sense → Process → Detect → Communicate → Alert”. The system also provides a foundation for an IoT-enabled mobile application through which boat operators, authorized monitoring personnel and rescue stakeholders can view alerts and status information. The estimated prototype development cost is ₹58,000, covering the base unit, Boat Hub, five wearable units, integration/testing and marine protection.

This project report presents the problem background, objectives, literature-oriented context, proposed architecture, components, working methodology, automatic rudder mechanism, software communication design, cost structure, feasibility, limitations, future scope and commercialization considerations.

INTRODUCTION

The marine fishing industry is one of the oldest and most significant occupations supporting the livelihoods of millions of people worldwide. In India, particularly in coastal states such as Tamil Nadu, Kerala, Andhra Pradesh, Karnataka, Goa, Odisha, West Bengal, and Gujarat, fishing serves as a major source of employment and economic development. Fishermen venture into the sea daily under challenging environmental conditions, where they encounter risks such as unpredictable weather, equipment failure, accidental falls into the water, communication loss, and unintentional crossing of international maritime boundaries. These hazards not only threaten their lives but also affect the socio-economic well-being of their families and coastal communities. Despite advancements in marine navigation and communication technologies, many small-scale fishermen continue to rely on conventional navigation methods and have limited access to modern safety equipment. Commercial marine safety devices are often expensive, bulky, or designed primarily for large vessels rather than individual fishermen. Consequently, there is a growing need for an affordable, intelligent, and wearable safety system capable of providing real-time monitoring, emergency communication, and life-saving assistance during marine emergencies. SMARTSEA BELT is proposed as an integrated wearable marine safety device that combines positioning, communication, health monitoring, and emergency flotation into a single system designed specifically for fishermen.

BACKGROUND OF THE STUDY

Marine accidents continue to pose significant challenges across coastal regions. Fishermen frequently travel several nautical miles from the shoreline in search of fish, exposing themselves to adverse weather, strong currents, limited communication networks, and navigational uncertainties. During emergencies, immediate rescue is often difficult due to delays in locating the victim or transmitting accurate position information.

The International Maritime Boundary Line (IMBL) is another critical concern for Indian fishermen. Limited awareness of exact maritime boundaries, navigation inaccuracies, and poor visibility may result in accidental border crossings. Such incidents can lead to legal complications, vessel seizure, financial losses, and emotional distress for families.

Current safety practices often depend on manual communication devices or standard life jackets. While these provide some level of protection, they do not continuously monitor the wearer's condition or automatically transmit emergency information. The integration of IoT, artificial intelligence, and satellite navigation technologies creates opportunities to develop smarter and more responsive wearable safety systems.

Over 100000 people are dying every day in india due to drowning or getting lost we wanted to make a solution for this with corporate social responsibility.

NEED FOR THE STUDY

The SMARTSEA BELT project has been developed to address several critical gaps in marine safety. Existing safety equipment generally performs only one function, such as flotation or communication. However, fishermen require a comprehensive solution capable of simultaneously monitoring health, tracking location,

providing emergency alerts, and assisting rescue operations.

The proposed system integrates multiple technologies into a single wearable platform, reducing dependence on separate devices while improving usability. The combination of NavIC GPS, LoRa communication, emergency alerts, and an inflatable flotation system aims to enhance survival chances during marine accidents and support faster rescue operations.

Furthermore, the project considers affordability and ease of deployment, making it suitable for adoption by coastal communities and fisheries departments. Sustainable to the environment and eco friendly Partnership focus with social responsibility,

PROBLEM STATEMENT

Fishing activities expose individuals to numerous life-threatening situations. Several accidents occur due to accidental crossing of the International Maritime Boundary Line, man-overboard incidents, delayed rescue response, medical emergencies, and inadequate communication infrastructure. Many existing marine safety devices either lack integrated monitoring capabilities or remain financially inaccessible to small-scale fishermen. There is a need for a compact, intelligent, and wearable marine safety solution capable of continuously monitoring the user's condition, providing accurate positioning, transmitting emergency information, and improving flotation during accidents. SMARTSEA BELT addresses these challenges by integrating multiple safety technologies into a single wearable device. Also, in India 10000+ die due to drowning or trekking, hiking, paraclimbing etc our product will help them to keep them safe and reduce drowning of kids while swimming or other activities.

OBJECTIVES

Primary Objective

To design and develop an GPS wearable marine safety system capable of improving fishermen safety through intelligent monitoring, emergency communication, and automatic flotation assistance and divert system.

Specific Objectives

- Develop a wearable safety belt suitable for marine environments.
- Provide accurate positioning using NavIC GPS.
- Enable long-range communication through LoRa technology.
- Monitor the user's health during fishing activities.
- Tracking them in a mobile application

- Generate emergency alerts during accidents.
- Integrate an inflatable flotation mechanism.
- Improve rescue response time.
- Reduce accidental IMBL crossing.
- Enhance affordability and accessibility.

SCOPE OF THE STUDY

The scope of SMARTSEA BELT includes the design, development, implementation, and evaluation of an integrated marine wearable safety system. The project focuses on fishermen operating in coastal waters and aims to improve their personal safety through wearable technology and every citizen who needs our product. The study covers hardware integration, software development, emergency communication, health monitoring, business feasibility, sustainability, and future commercialization. Although developed primarily for fishermen, the proposed technology may also be adapted for marine rescue teams, coastal security personnel, aquaculture workers, recreational boating activities, and water sports applications.

Research Methodology

The research methodology adopted for SMARTSEA BELT consists of the following stages:

1. Problem Identification
2. Literature Survey
3. Requirement Analysis
4. System Design
5. Component Selection
6. Hardware Development
7. Software Development
8. Prototype Fabrication
9. Testing and Validation
10. Business Analysis
11. Sustainability Assessment
12. Documentation

Each stage contributes to the systematic development and evaluation of the proposed wearable safety device.

SIGNIFICANCE OF THE STUDY

The SMARTSEA BELT project contributes to marine safety by introducing an affordable and integrated wearable system designed specifically for fishermen. Unlike conventional safety devices, the proposed system combines positioning, communication, health monitoring, and flotation support within a single compact platform. The project also demonstrates the practical application of IoT technologies, intelligent sensing, and wearable computing in addressing real-world challenges faced by coastal communities.

INNOVATION

The novelty of SMARTSEA BELT lies in the integration of multiple technologies into one wearable device.

Key innovative features include:

- NavIC-based marine navigation.
- LoRa-based long-range communication.
- Health monitoring.
- One-touch SOS emergency alert.
- 30 degree divert system
- Automatic flotation assistance.
- AI supported emergency management.
- Expandable platform for future satellite communication and predictive analytics.

This integrated approach seeks to improve usability while reducing equipment complexity for fishermen.

Reference Data sets that inspired us to implement this idea.

Protecting of Fishermen on Indian Maritime Boundaries

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Abstract - The Tamil Nadu fishermen even today invoke the historical rights and routinely stray into the International Maritime Boundary Line (IMBL) for fishing. This has led to apprehension by the Sri Lankan Navy and in some cases even to shooting or arrest the particular Fishermen. This leads to loss in the both humans as well as their economic incomes. The reasons are the lack of Global Positioning System (GPS) in mechanised boats and a chain of Automatic Identification System (AIS) stations along the Tamil Nadu coast areas. So we are going to produce a system which uses several modules to protect the fishermen. This concept makes use of effective ways in the embedded systems and some basic principles in physics. The three major parts which makes the Indian coastal navy to respond for the issues, fishermen can protect them self from the Lankan navy and automatic warning for the unaware fishermen.

Keyword: *Automatic Identification System , International Maritime Boundary Line, Global Positioning System.*

I. INTRODUCTION

The project induces the new methodology for saving the fishermen's valuable life and their properties from the srilanka navy. This system makes use of arduino which is open source embedded kit available for much major purposes. Arduino is an open-source single-board microcontroller, descendant of the open-source Wiring platform, designed to make the process of using electronics in multidisciplinary projects more accessible. The hardware consists of a simple open hardware design for the Arduino board with an Atmel AVR processor and on-board input/output support. The software consists of a standard programming language compiler and the boot that runs on the board. The GPS72h is the equipment used normally by all the fishermen in the navigation in sea for the purpose of identification. It is a Satellite Navigation based Equipment also GPS 72H acquires satellite signals quickly and tracks your location in challenging conditions, such as heavy tree cover or deep canyons. The GPS 72 H floats in water and its IPX7 waterproof to withstand the accidental dunk or splash in the lake. Whether on water or land, the GPS 72H can save up to 500 of your favourite places in memory and point you to your destination (no street or terrain maps). The GPS 72H is NMEA 0183 compatible. To transfer data to your device, GPS 72H easily connects to your computer via USB. If you're into boating, hunting or fishing, you can use the 72H's built-in celestial data, which includes sunrise/sunset times and

a hunting and fishing calendar. The important system is RADAR detector used in our proposal. Radar detectors use a super heterodyne receiver to detect these electromagnetic emissions from the radar gun, and raise an alarm to notify the motorist when a transmission is detected. False alarms can occur however due to the large number of devices, such as automatic door openers (such as the ones at supermarkets), that operate in the same part of the electromagnetic spectrum as radar guns.

Objective: Protect the Tamilnadu fishermen from crossing the Indian-Srilanka maritime boundaries and thus save their life, economic status of fishermen.

II. EXISTING SYSTEM

At present, there are few existing systems which help to identify the current position of the boats/ships using GPS/RADAR Navigation system and view them in an electronic map. This provides the fastest and most accurate method for mariners to navigate, measure speed, and determines location. This enables increased levels of safety and efficiency for mariners worldwide and accurate position, speed and heading are needed to ensure the vessel reach its destination safely. The accurate position information becomes even more critical as the vessel departs from or arrives in port and a person have to watch the system for the detecting the malpractice of the boats. In this case there may be manual error to find the boats crossing our boundaries. Also the information regarding the boundary crossing boats is have to pass to the higher official, coastal guards by manually. This will also give a time to trace and warn the boats.

Another system is GPS72H by the GARMIN which is mostly used by the fishermen is battery based power supply it stands for 18 hours but the fishermen may be unfortunately missing their backup batteries will lead to danger . Also another problem is this GPS72H also to be manually watched to identify whether they crossed the boundary. There is no indication for the fishermen whether they are traced by the foreign navy.

According to the project of Mr.Naveen from St.Joseph College of engineering Chennai .A equipment which can stops the fuel connection to the engine of the fishermen vessel when they tend to cross the boundary. They will give a chance to restart the engine with reverse motion if not they fuel supply is permanently disconnected. This system leads to make the vessels to trap on nearer to boundaries and which may turn too caught by the foreign navy. Also this system can be applicable only to the inboard engine and not for the outboard engine based vessels.

III. PROPOSED SYSTEM

The proposed work consists of three major modules in order avoid all the shortcomings of the existing system. The modules are Vessel tracking module, RADAR identification module, up gradation in GPS72H.

The first module Vessel tracking has the theme of tracking the vessels by using the RADAR Navigation used in the Indian coastal vessels. In the existing system of RADAR Navigation the Indian coastal vessels using the Rajendra Prasad Radar which is a passive Phased Array Radar developed by the Indian DRDO. It is multifunction radar, capable of surveillance, tracking and engaging low radar cross section targets. It is the heart of the Akash Surface-to-air missile system and is the primary fire control sensor for an Akash battery. This navigation shows the status of the vessels in an electronic map. The electronic map should be properly and periodically watched by men who could be creating a manual error to miss their concentration towards the vessels tracked. In order to avoid this we are going to make a system which would send the data of electronic map i.e. the position of each vessel which is saved in the RADAR system to a normal PC which runs our VB program in infinite times. Each time the program receives the value checks with the constant boundary values of the Indian maritime boundary (The values are expressed instead of the latitude and longitude position of the particular point). The result of the condition checking will produce the output as an alarm signal from the speaker connected to the PC.

After the alarm is blown the navy guards can easily track the person's vessels who are trying to malpractice. Another advantage added with this system is if the navy coast has not responded for a certain fixed time a message in the form of text containing the details of the position of the vessels which are going to cross the boundary is pushed to the cell phone of higher officials. This will grab the attention of the higher officials to the fishermen when they are in danger even though the respected navy coast has not responded. This vessel tracking system is the complete updating of a new system (PC based warning) with existing RADAR Navigation in the coastal vessels.

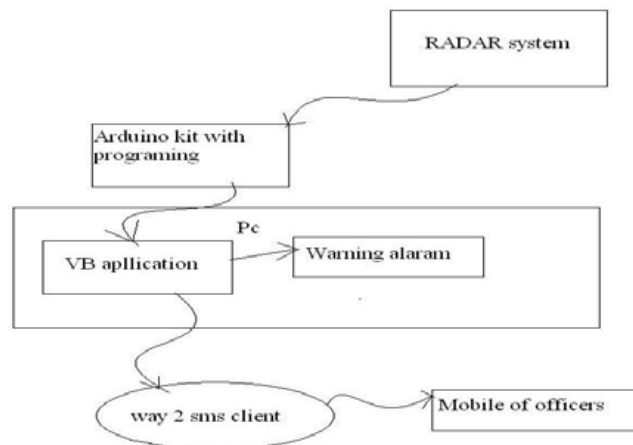


Figure1:Vessels Tracking Module

The second important system is RADAR identification module. In this equipment called as RADAR Detector is used. Radar detectors use a super heterodyne receiver to detect these electromagnetic emissions from the radar gun, and raise an alarm to notify the motorist when a transmission is detected. False alarms can occur however due to the large number of devices, such as automatic door openers (such as the ones at supermarkets), that operate in the same part of the electromagnetic spectrum as radar guns. Using this principle the same

receiver is going to mount in the boats of Indian fishermen. The output i.e. the indication of the RADAR will gives the alarm which may understood as that they are traced by the foreign navy so the fishermen can easily navigate from that place before they caught by the foreign navy. In normally the fishermen are visually indentifying the foreign navy which may not give a time to escape themselves from the foreign navy. But this RADAR detector is very help full for the fishermen to protect themselves.

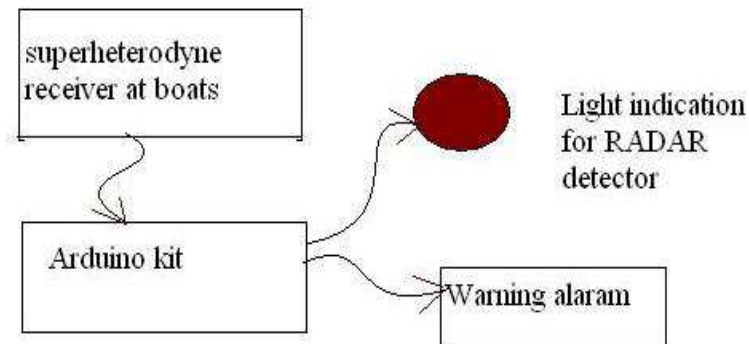


Figure 2: Radar Detection System

The third module is the Updating GPS72H. The Garmin GPS 72H is a handheld, waterproof GPS receiver that floats. It measures 2.7 x 6.2 x 1.2 inches and has a 1.6 x 2.2 inch high contrast display with a resolution of 120 x 160 pixels. The display is capable of displaying 4 levels of gray for improved readability. The unit operates up to 18 hours on 2 AA batteries, and weighs only 7.7 ounces with batteries. The unit is waterproof to IPX7 standards and will float if dropped in water. It includes serial and USB interfaces for PC connection, can save up to 500 waypoints, 50 routes and a 2,048 point track log with up to 10 tracks. The unit has a 9-button user interface that is suitable for one-handed operation however the GPS 72H has no street or terrain map capability. As we know that the GPS72H is the battery based power supply. To avoid the disadvantage of GPS72h which mentioned earlier a solar panel based charger is to be upgraded with the equipment in order to make the equipment not to lose their power and another major updating is the GPS72H receiver equipment is to be embedded with the Arduino kit for the process of automatic warning system. The Automatic warning system includes the saved waypoints values of the display in GPS72H are checked with the constant maritime boundary values (The values are expressed instead of the latitude and longitude position of the particular point). If the results of the comparison shows the boats/vessels is beyond the Danger area (area which is 0.5 km nearer to maritime boundary) the alarm attached to the arduino kit will supplied with the 5V and make it to blow until the fishermen stops that. This makes the illiterate fishermen to understand that they are at the danger area which makes them to fishing at the safer zone.

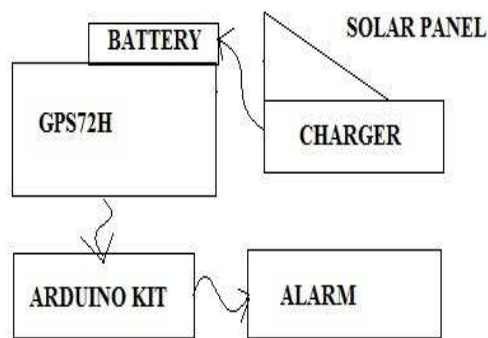


Figure 3: Updating GPS72H

These three modules are comprised of many programming parts. The vb code and arduino kit code for the vessel tracking module.

ARDUNIO CODING

```
SoftwareSerial port
mySerial.begin(4800);
mySerial.println("Hello, world?")
}
int i;
void loop()
{
  i=analogRead(3);

  if (mySerial.available())
    Serial.write(mySerial.read());
  if (Serial.available())
    mySerial.write(Serial.read());
}
```

Figure 4: Arduino Programming

The shown programming is uploaded in to the Arduino kit in order to read the analog value from the RADAR Navigation system and write those values into the serial port of the pc which is further used for the condition checking in vessel tracking module.

VB CODE FOR SERIAL READING

```

Public Static Sub ShowData(TextShow As Control, Data As
    Const SpeedBaud = 16000
    Dim lngSize As Long, X
    lngSize = Len(TextShow.Text)
    If TermSize > SpeedBaud Then
        TextShow.Text = Mid$(TextShow.Text, 4097)
        lngSize = Len(TextShow.Text)
    End If
    TextShow.SelStart = SpeedBaud
    Do
        X = InStr(Data, Chr$(8))
        If X Then
            If X = 1 Then
                TextShow.SelStart = SpeedBaud - 1
                TextShow.SelLength = 1
                Data = Mid$(Data, X + 1)
            Else
                Data = Left$(Data, X - 2) & Mid$(Data, X + 1)
            End If
        End If
    Loop While X
    Do
        X = InStr(Data, Chr$(10))
        If X Then
            Data = Left$(Data, X - 1) & Mid$(Data, X + 1)
        End If
    Loop While X
    X = 1
    Do
        X = InStr(X, Data, Chr$(13))
        If X Then
            Data = Left$(Data, X) & Chr$(10) &
            Mid$(Data, X + 1)
            X = X + 1
        End If
    Loop While X
    TextShow.SelText = Data
    TextShow.SelStart = Len(TextShow.Text)
End Sub

```

Figure 5: vb Code for Read Value from Serial Port

In case of RADAR identification module these code is slightly changed i.e. the serial port written is nullified instead of that the pin is supplied with high voltage which is connected to the alarm. The following codes are the codes for reading the written value by Arduino kit at the serial port. After this reading value at the vessel tracking module the values read is compared with the constant boundary values.

VB CODE FOR COMPARSION

```

If (a > 12 And b > 12) Then
    WindowsMediaPlayer1.URL =
    "E:\ALARAM.mp3"
    WindowsMediaPlayer1.Controls.play
    ym = MsgBox("Click OK to Proceed",
    vbOKCancel, "Startup Menu")
    If (ym = 1) Then
        WindowsMediaPlayer1.Controls.stop
    End If
End If

```

Figure 6: Alarm Code at pc

In the RADAR Detector module the same Arduino kit coding mentioned in Fig.4 is used to read the values from superhetrodyne receiver. Also in the Updating GPS72H module contain this same code for reading values from that equipment and both the module contain the checking the values with constant value then supplying 5v to the pin connected to the alarm. The constant values referred in the all the modules can be changed according to the maritime boundary values.

Boundary Locations

The maritime boundary between Sri Lanka and India in the Gulf of Mannar shall be arcs of great circles between following positions, in the sequence given below, defined by latitude and longitude. The maritime boundary between Sri Lanka and India in the Bay of Bengal shall be arcs of great circles between the following positions, in the sequence given below, defined by latitude and longitude. Maritime boundary in Bay of Bengal

POSTIONS LATITUDE LONGITUDE

Position 1 09° 06'.0 N 79° 32'.0 E
Position 2 09° 00'.0 N 79° 31'.3 E
Position 3 08° 53'.8 N 79° 29'.3 E
Position 4 08° 40'.0 N 79° 18'.2 N
Position 5 08° 37'.2 N 79° 13'.0 E
Position 6 08° 31'.2 N 79° 04'.7 E
Position 7 08° 22'.2 N 78° 55'.4 E
Position 8 08° 12'.2 N 78° 53'.7 E
Position 9 07° 35'.3 N 78° 45'.7 E
Position 10 07° 21'.0 N 78° 38'.8 E
Position 11 06° 30'.8 N 78° 12'.2 E
Position 12 05° 53'.9 N 77° 50'.7 E
Position 13 05° 00'.0 N 77° 10'.6 E

POSITIONS LATITUDE LONGITUDE

Position 1 10° 05'.0 N 80° 03'.0 E
Position 2 10° 05'.8 N 80° 05'.0 E
Position 3 10° 08'.4 N 80° 09'.5 E
Position 4 10° 33' 0 N 80° 46'.0 E
Position 5 10° 41'.7 N 81° 02'.5 E
Position 6 11° 02'.7 N 81° 56'.0 E
Position 7 11° 16'.0 N 82° 24'.4 E

The boundary between India and Sri Lanka in the waters from Adam's Bridge to Palk Strait shall be arcs of Great Circles between the following positions, in the sequence given below, defined by latitude and longitude.

POINT LATITUDE NORTH LONGITUDE EAST

1 10° 05' 80° 03'

2 09° 57' 79° 35'

3 09° 40'.15 79° 22'.60

4 09° 21'.80 79° 30'.70

5 09° 13' 79° 32'

6 09° 06' 79° 32'

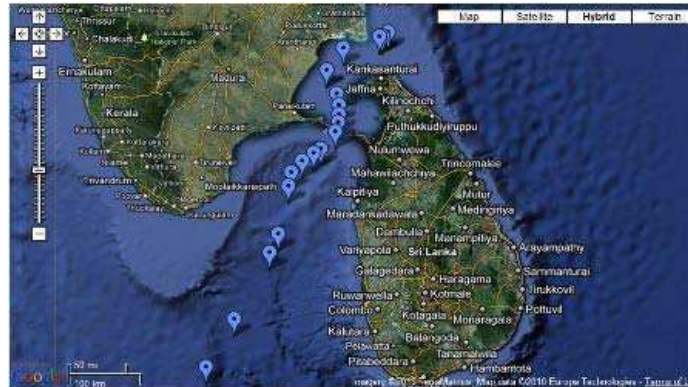


Figure7:Indian Srinlanka maritime boundaries

IV. CONCLUSION

According to our concepts the every fisherman can be traced automatically by Indian costal guards and also they can be safer and they can protect themselves from the foreign navy which leads to a pleasurable life for them.

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IoT Based Border Alert and Secured System for Fisherman

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Abstract— An embedded system which protects the fishermen by notifying the country border to them by using Global Positioning System (GPS) and Global system for mobile communication (GSM). Using GPS, we can find the current latitude and longitude values and is sent to the microcontroller unit. Then the controller unit finds the current location by comparing the present latitude and longitudinal values with the predefined value. Then from the result of the comparison, this system aware the fishermen that they are about to reach the nautical border. Looking at this operation from an IoT (Internet of Things) lens, and taking account of the latest technology, it seems necessary and desirable to integrate the data sources to make sure that the perspective that decision makers need is based on a complete picture of the situation – not just whether the traveler or a vehicle is on the watch list. This paper looks at some of the issues and opportunities that appear to be worth considering. It is necessarily a cursory look presented in the hope that it will be of use as the thought process for such a project starts. These are strictly personal views of the authors who have no connection with any law-enforcement agency. If the boat enters the zone nearer to the restricted zone the alarm will turn on and the speed of the boat engine automatically gets controlled by the concern authority and once it touches the restricted zone and its pulled back to the non-restricted area.

Index Terms— Global Positioning System (GPS), Global system for mobile communication (GSM), nautical border

I. INTRODUCTION

In this modern, fast moving and insecure world, it has become a basic necessity to be aware of one's safety. Maximum risks occur for fishermen in situations where they travel on a boat for fishing. The Tamil Nadu fishermen even today invoke the historical rights and routinely stay into the International Maritime Boundary Line (IMBL) for fishing. From Tamil Nadu about 18,000 boats of different kinds conduct fishing along the India-Sri Lanka maritime border. But by accidentally crossing the border without knowledge, they get shot by the Lankan navy. This leads to loss in the both humans as well as their economic incomes. We have developed a system which eliminates such problems and saves the lives of the fishermen. In some situations they should not move after some point and they should not enter into other countries area. There is a real necessity in designing a system that can track the vehicle and send the information about the vehicle to the concerned person

And alert the fishermen also. The project aims in designing a system which is capable of alerting the fisherman when their boat is crossing the country border and also displays the alert message on a Liquid Crystal Display (LCD). GPS based border alert system gives a best solution for this problem, whenever the fisherman was about to reach the boundary he can have a voice based alert and also displays the speed of the boat on LCD. So that he can go back from that point onwards.

II. LITERATURE SURVEY

There are several projects undertaken and various methods proposed for border alerting for small boats. Various technologies has been used to implement this keeping safety of fishermen as motto. Following are the few papers which propose the idea of border alerting measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

1) Advanced navigator and monitoring system for fisherman

By Supriya Lohar, The lives of Indian fishermen is very important. Thousands of Tamil fishermen have been arrested, shot by the Sri Lankan government. Fishermen not having the knowledge of boundary, enter into the Sri Lankan boundary. Once entered into the boundary, they get caught or shot by the government. Hence to save the lives of fishermen we have designed the system which will help the fisherman not to navigate beyond the boundary by using the (Global Positioning System) GPS and (Global System for Mobile Communication) GSM. The main objective of this project is to help the fisherman not to navigate beyond the country's border. Using the GPS we can find out the current location of ship or vessel. Here we are using three boundary locations for which we are giving alert to the fisherman.

2) Secure and efficient border crossing indicator application in coastal area

It had been proposed by R. Sangeetha, P.V.Jothikantham and T. Senthil kumar Maritime surveillance is carried out by (and on behalf) of national authorities mainly to identify and deter (a) infringements to regulations and (b) security and safety threats. (These categories overlap and include law enforcement and compliance monitoring.) Surveillance carried out for the

different domains, such as fisheries protection, environmental protection, maritime transport safety, border control, etc., in most cases falls under (a) because it is done on the basis of laws and regulations that govern these domains. Surveillance is a key element to exercise national sovereignty at sea.

3) Intelligent boundary alert system (IBAS) using GPS

GPS had been proposed by C. Sheeba thangapushpam. This system helps the fishermen in maritime navigation. The system uses a GPS which continuously receiving signals from the satellite and provide the current position of the boat based on the latitude and longitude data. ARM processor is already fetched details of the latitude and longitude of the maritime boundary. Comparison is done by the processor with stored data and current position of the boat, and it generates the alarm signal whenever the boat crosses the border. They used wireless sensor network to transmit the message to the base station, there they monitors the boat in the sea. This system provides an indication to both fisherman and to coastal guard [1].

4) Implementation of GPS based security system for safe navigation of fisherman auto boat

It was proposed by C. Vinothkumar and B. Arunkumar, that also uses GPS technology for navigation and vessel tracking purposes Using microcontroller, the stored border data between India and Srilanka is being compared with the current location details of the boat, and then alarm signal is being generated when the vessels crosses the border. Simply the message will be transmitted to the base station. In addition, some sensor is used to detect the natural calamities for sea way travel. The ultrasonic sensor is used for the detection of the iceberg, and MEMS is used for tsunami detection. In addition to this weather forecasting report can also be obtained with the help of temperature and humidity sensor [2].

5) IOT based nautical monitoring system

It had been proposed by Kannan.K.R, Hakkem.B, Vignavi.K, Vinoth.P., Mohammed Noorul Karim.M, Rajbab The current latitude and longitude is known to both fishermen and coastal guards. The border is identified by comparing the current values with the original values and the message is sends through Wi-Fi sensor using IOT. Single antenna is used. This helps in continuous updating of the information at instantly. The low earth orbit is used to provide the connectivity without gap. This also helps in retrieving the missed messages. By using the electronic map, the navigation path can be identified. Thus it saves the lives of the fisherman and alerts the base station to provide help.

6) Automatic border alert system for fishermen using GPS and GSM techniques

It had been proposed by S. Ranjith, Shreyas, K. Pradeep Kumar, R. Karthik. The proposed system is used to detect the border of the country through the specified longitude and latitude of the position, not only between Sri Lanka and India but all over the world. The particular layer level i.e. border can be predefined and this can be stored in microcontroller memory.

The current value is compared with predefined values and if these values are same, immediately the particular operation will be done i.e, the microcontroller gives instruction to the alarm to buzzer. It also uses a message transmitter to send message to the base station which monitors the boats in the sea.

7) Location based services using android

It had been proposed by Archana Gupta, Mohammed Abdul Qadeer, Sandeep Kumar From the available GIS processing tools in Android we can realize all three types of LBS services as a mobile can be configured as a server and for that we can also use the SQLite database to store information as android also supports this technology. We can configure the two mobiles to provide peer-peer LBS services through SMS or MMS with the use of appropriate intents.

8) Alert fishers: a lifesaving mobile application

Mobile Application which can save time, effort and money and life by giving tips and ways to reduce the arrest of fisher beyond boundaries that is designed and implemented. The collected information contains all the details related to countries boundary from the boundary defining department of India and Sri Lanka, and grouped them into an easy and compact application for the benefit of the each and every fisherman.

III. WORKING PRINCIPLE

The GPS Modem will continuously give the signal which determines the latitude and longitude and indicates the position of the fishermen to them. Then it gives the output which gets read and displayed in the LCD. The same data is sent to the mobile of the fisherman and simultaneously the same data is sent to the Sea border security. An EEPROM is used to store the data, received by GPS receiver. The hardware which interfaces with microcontroller are LCD display, GSM modem and GPS Receiver. GPS (Global Positioning System) is increasingly being used for a wide range of applications. It provides reliable positioning, navigation, and timing services to worldwide users on a continuous basis in all weather, day and night, anywhere on or near the Earth. 28 satellites inclined at 55° to the equator orbit the Earth every 11 hours and 58minutes at a height of 20,180 km on 6 different orbital Lanes and each one of these satellites has up to four atomic clocks on board.



Fig. 1. Gps signal transmission

All we require is an accurate clock. By comparing the arrival

time of the satellite signal with the onboard clock time, at which the signal was emitted, the latitude and longitudinal degree of the boat's location is determined. The current design is an embedded application, which will continuously monitor a moving Boat and once the boat goes beyond the level of the defined layer the particular operation will be done. For doing so an AT89c51microcontroller is interfaced serially to a GSM MODEM AND GPS receiver

IV. EXISTING SYSTEM

There are many disadvantages in the existing system like the lack of awareness of border which causes accident. Although, there are borders defined but it is a blue eyed vision of people of country, it is important to pay their attention.

1) GPS 72h

The GPS 72h equipment is used for navigation in the sea [2]. It provides the fastest and most accurate method for mariners to navigate, measure speed and determine location.



Fig. 2. GPS 72h

This system enables increased levels of safety and efficiency. It ensures that the ship reaches its destination safely. The accurate position information becomes even more critical as the vessel departs from or arrives in port

2) Distress alert transmitter (DAT)

This device was jointly developed by the Indian Coast Guard and the Indian Space Research Organization (ISRO) for fishermen [5]. INSAT based Distress Alert Transmitter (DAT) is used for transmitting conditions along with current location of the vessel to the central hub station via transponders.

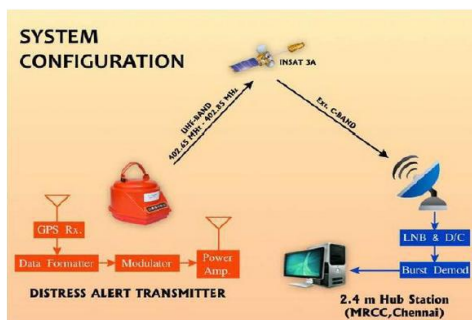


Fig. 3. DAT system configuration

Transponder can be easily installed in boats or other vehicles. In case of emergency, fisherman just needs to switch on the DAT unit. If Fisherman needs to transmit emergency message

such as fire on vessel or medical emergency, the message can be selected by pressing the corresponding switch which is provided in the system. After pressing the appropriate switch DAT emits the signal. The message is transmitted through the satellite to the coast guard. DAT combines the message signal along with the current location of the vessel to central station. Features: Low Cost and therefore affordable to fishing community. In-built GPS to give position and time information.

3) Vessel traffic management system

The System includes integrated RADAR, Automatic Identification System (AIS), Radio Direction Finders (RDF), Metrological and Hydrological Sensors, Microwave links as well as VOIP based Very High Frequency (VHF) radio system [5] All maritime traffic is recorded and movements are stored in a database.

4) Automatic identification system (AIS)

In a system transponders are installed on vessels which are used to transmitting signals [5]. The data is sent from each vessel which contains current position of the vessel as well as movement information such as speed, longitude and latitude No alarming and warning for obstacles and tsunamis. No security system is present which causes loss of life.

5) Drawback of the existing system

The drawback of the all existing systems is that all systems give current location of the vessel and traffic monitoring, some systems transmit the emergency message to the base station. But, none of the existing system is give an alert to the fisherman about the boundary locations. Hence to save the life of fisherman we have proposed a system which gives alert to the fisherman about the boundary locations

V. PROPOSING SYSTEM

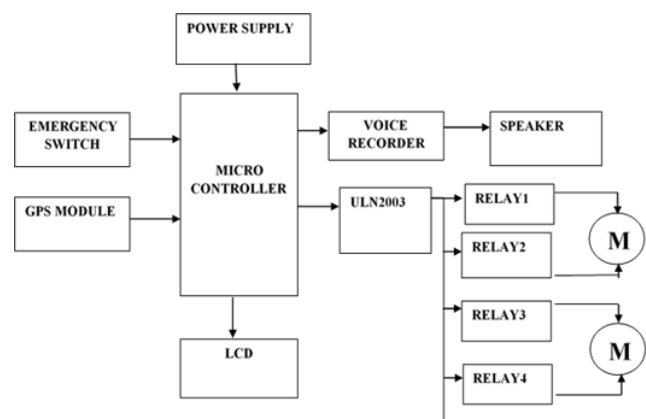


Fig. 4. Proposed main architecture

In the boat module microcontroller is used to control all the activities at the boat. This microcontroller is programmed in such a way as to it continuously track the boat and send GPS signals to the system at coastal guards. So once the boat crosses the safe zone the microcontroller sends a signal to stop engine of the boat and engine will be stopped. And the boat is controlled by server unit. It also helps to display various

messages on LCD and voice module based on signals. GPS module with antenna is used to track the location of the boat at every instance of time. This location is sent to the base station with the help of microcontroller. LCD is used to display various alert messages for the boat at various situations and along with the LCD a voice module is also used which alert the people at the boats. The safe zone set by the control unit using keypad. The safe zone value is before the border this is useful to give the alert before crossing. The control unit receive the signal if engine is stopped or some critical situation the control unit control the boat based on the fishermen request.

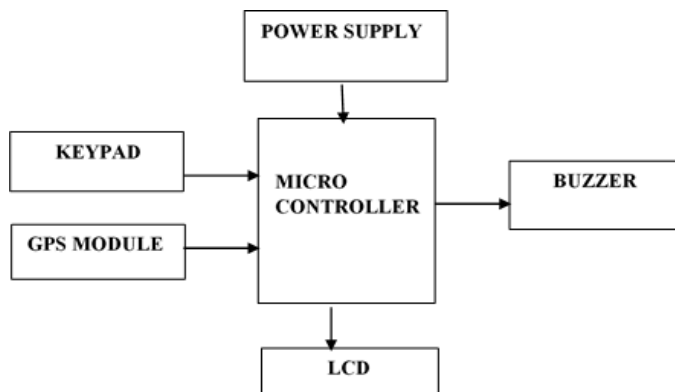


Fig. 5. Proposed Control Architecture

VI. PROPOSED SYSTEM FLOW DIAGRAM

The GPS receiver receives the signal and converts it into desired data message [2]. The data is sent to the microcontroller and controller extracts the longitude and latitude information from the data. Positions get compared with the predefined locations. If the location is matched then alarm is generated and message is transmitted to the base station and also to the fisherman

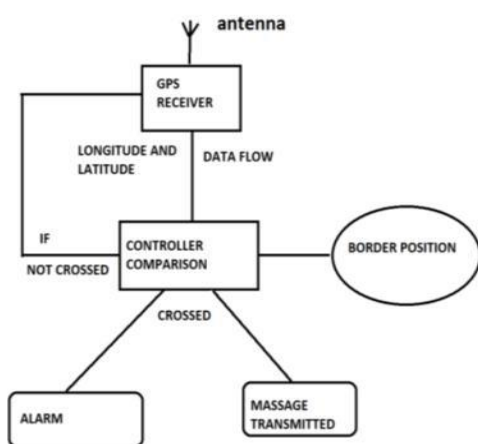


Fig. 6. Proposed System

VII. TECHNOLOGY USED FOR THE SYSTEM

1) Global positioning system (GPS)

It is a space based satellite navigation system. GPS gives timing and location information in any weather conditions. GPS

works in all climate conditions. GPS devices are used in military, marine and consumer product applications.

2) GPS Accuracy

The accuracy of the GPS depends on the type of receiver. Most hand-held GPS units have about 10-20 meters accuracy.

3) GSM M GSM

Module is used for the message transmission. If the fisherman is about to cross the final boundary at that time GSM gets connected to the controller and sends an alert message to the fisherman and base station.

4) Controller

GPS is continuously connected to the controller unit. The controller receives the data from the GPS. The data contains latitude and longitude of current location of the vessel. We have already stored the boundary locations in the controller unit. When controller receives data from the GPS, controller starts comparing current location with the already stored locations. When the locations get matched then controller performs the respected operation such as it gives the instruction to the buzzer to alert the fisherman and also if fisherman neglect the alert and moves further then the controller starts comparing current location with already stored location and if the location gets matched then controller unit gives instruction to the LCD display an alert message to the fisherman.

VIII. CONTROLLER PIC MICROCONTROLLERS

(Programmable Interface Controllers), are electronic circuits that can be programmed to carry out a vast range of tasks They are found in most electronic devices such as alarm systems, computer control systems, phones, in fact almost any electronic device PIC Microcontrollers are relatively cheap and can be bought as pre-built circuits or as kits that can be assembled by the user. The data received contains many details along with latitude and longitude. The current positions are compared with already stored latitude and longitude of countries boundary locations. At first the latitude is compared with stored latitude which identifies if the current position is located near to the boundary. If the latitude matches then the adjacent latitudes and longitudes of the present latitude is retrieved from the microcontroller. The current position received from GPS is stored as S1 (latitude), S2 (longitude). The latitude S1 is compared with stored latitudes. If latitude match, then adjacent latitude and longitudes (X1, Y1 and X2, Y2) are retrieved from stored table and substituted in the equation given below: $(Y-Y1)/(Y2-Y1) = (X-X1)/(X2-X1)$ By simplification, we get $ax + by = c$ Now, S1 and S2 are substituted in above equation of line. Positions Latitude Longitude Position 1 12° 05'.0 N 82° 03'.0 E Position 2 12° 05'.8 N 82° 05'.0 E Position 3 12° 08'.4 N 82° 09'.5 E Position 4 12° 33'.0 N 82° 46'.0 E

IX. LAYER CONSTRUCTION AND CONSEQUENCE

The boundary points should be stored in microcontroller and the computation is done in microcontroller with these points.

Thus vessel crossing the border is being calculated. Boat Position and Navigation System contains

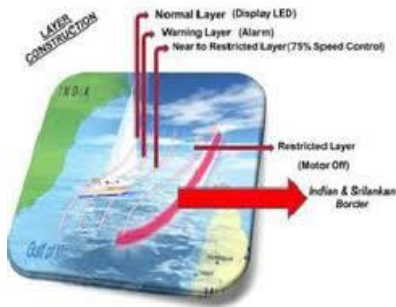


Fig. 7. Layer construction

1. Layer1: Green LED indication
2. Layer2: Red LED indication
3. Layer3: Alarm indication & speed control
4. Layer4: Engine off

X. CALCULATING THE POSITIONS

Because GPS receivers do not have atomic clocks, there is a great deal of uncertainty when measuring the size of the Spheres. All possible distances to the satellite are located on the circumference of the circle. If the position above the satellites is excluded, the location of the receiver is at the exact point Where the three circles intersect beneath the satellites. Because the relative size of the spheres is known, there is only one possible point where they can intersect.

XI. MARITIME BOUNDARY BETWEEN INDIAN AND SRI-LANKA

The boundary points are marked above. These points should be stored in microcontroller. The computation is done in microcontroller with these points. Thus vessel crossing the border is being calculated. Consequence Boat Position and Navigation System contains,

1. Location 1: buzzer indication
2. Location 2: motor speed control indication.
3. Location 3: motor stops.
4. Location 4: final verification

XII. EXPERIMENTAL RESULTS

Border alert system for fishermen is used to detect the boundary location and warn the fishermen in danger situations.



Fig. 8. Alert border

It not only finds the GPS value, but also compares with the stored value in the microcontroller, and makes a decision as to whether the fishermen is in the warning range or not.

XIII. FUTURE SCOPE

1. We can use the EEPROM to store the previous Navigating Positions up to 256 locations.
2. We can navigate up ton number of locations by increasing the memory of EEPROM.
3. We can reduce the size of the kit by using GPS+GSM on the same module of GPS navigator.
4. We can increase the accuracy up to 3m by increasing the cost of the GPS receivers.

XIV. BENEFITS

1. The hijack of the ship by the pirates can be eradicated.
2. The lost ship wrecks due to natural calamities can be identified
3. By keeping the kits in the entire boats and by knowing the locations of all the boats we can use our kit to assist the traffic.
4. In case of any accident on the sea. It can be detected by the system and the accident location of the boat is sent to the rescue team.

XV. CONCLUSION

- Risk of fishermen in border line due to unknowingly crossed the border could be reduced by this system.
- Thus saving their lives and providing good relationship with the neighboring countries. Also, the piracy of ship can be easily brought under control.
- It is not easy to find the border range as in land for anyone in marine region. When they crossed the border limit they have to pay penalty or got arrested by neighbor country navy guards.
- This project helps the fishermen to guide by navigation and alerting them when reached the border limit.
- By using emergency switch fishermen can be navigated back if they have lost their way in the sea or can be sent help if they are in danger.
- This concept of vessels tracking may enlarge to individual's vehicle security, with the GSM transmitter.

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A Distributed Edge-Intelligent IoT Network Framework for Hybrid Vision and Physiological Drowning Detection

Abstract

Drowning is a significant cause of accidental death across the globe, especially among children and those in unsupervised water bodies. Current detection methods are primarily based on individual sensors or vision based methods, which do not provide good opportunities to intervene in time in case of an emergency. This research introduces Aqua Guard, an Internet of Things (IoT) enabled real time drowning detection and automated rescue system that addresses the constraints of water bodies. The system incorporates physiological sensing using an ESP32 microcontroller, computer vision and behavioral analysis based on You Only Look Once version 8 (YOLO v8), and low latency communication via the Message Queuing Telemetry Transport (MQTT) protocol. The rescue mechanism is based on a stepper motor and a grid based localization module, which is interactive and allows a targeted and fast response. The experimental analysis showed a mean Average Precision (mAP) of 91.78, and the detection performance remained consistent across different lighting and occlusion conditions. Besides technical reliability, Aqua Guard is a privacy conscious, scalable, and affordable solution suitable for smart pools, community aquatic facilities, and other IoT related safety systems. The system is an essential advancement in automated drowning prevention.

Keywords: Distributed sensing systems; Internet of Things (IoT); MQTT-based messaging; Edge computing; Computer vision at the edge; YOLOv8.

I. Introduction

The rapid development of the Internet of Things (IoT) has transformed healthcare, safety, and environmental monitoring, enabling continuous data collection, intelligent analysis, and real time decision making. Wearable sensors and edge computing systems are now part of modern healthcare monitoring systems. These can continuously measure vital physiological parameters such as heart rate, blood oxygen saturation (SpO₂), respiratory rate, and physical activity. The

developments have significantly contributed to the detection and management of possible health risks at an early stage [19].

Although these technologies have been developed, IoT technologies are still applied sparingly in the field of aquatic safety. The application of physiological surveillance and smart sensing to water sources would enable more effective, proactive safety interventions and support drowning prevention.

Drowning is a serious worldwide social health issue. The World Health Organization (WHO) acknowledges that it is the third leading cause of accidental deaths related to injuries, claiming about 236,000 lives every year. The WHO Guidelines on Recreational Water Safety reiterate that high level surveillance and monitoring mechanisms are necessary to overcome the limitations of human observation in aquatic settings. Pediatric research has also pointed out that drowning incidents are usually nonverbal and within a few minutes, particularly in children and young adults [4]. Poor environmental conditions, including glare, overcrowding, and ineffective lighting, make it even harder to ensure timely detection. Even though international standards such as the International Swimming Pool and Spa Code (ISPSA) provide structural and safety requirements, traditional preventive measures, including lifeguard supervision, alarms, and flotation devices, are limited by human attention and response time.

Recent technological work has discussed AI based drowning detection solutions based on both computer vision and wearable sensing. Vision based systems identify irregular swimming activities, extended submersion, or inactivity using cameras and machine learning algorithms. Nonetheless, these techniques tend to be sensitive to lighting, reflections, occlusions, and privacy is an issue. Instead, wearable based still use physiological sensors to directly measure vital signs and movement patterns, although they have problems with waterproofing, user compliance, sensor drift and calibration accuracy [9], [13], [23].

This research presents a drowning detector and autonomous rescue system, Aqua Guard, a hybrid IoT based solution that combines physiological sensors, real time vision analytics, and automated mechanical controls to address these limitations. Aqua Guard will be configured to detect and save drowning victims quickly, reliably, and without being intrusive, reducing dependence on human supervision.

The system is composed of two major subsystems:

- **Wearable Unit:** It is based on the SP32 microcontroller, and it combines the Max30102 optical heart rate and SpO2 sensor and the BMP180 pressure sensor to estimate depth. The sensor data are wirelessly relayed to a central MQTT broker, where real time processing and classification are performed.
- **Vision Unit:** It functions by using a deep learning model, based on YOLOv8, to achieve behavioral detection and grid based localization of swimmer.

When a drowning event is confirmed by one of the subsystems, an autonomous net based rescue mechanism is activated using an Arduino controlled stepper motor. The system's architecture focuses on low latency, modular scalability, and energy efficiency, applicable in both public and private pool environments.

The key findings of the study can be discussed in the following way:

- **Multi modal Drowning Detection Architecture Hybrid:** The framework of the hybrid multi modal drowning detection system, which combines physiological and computer vision to minimize false positives and increase reliability time.
- **Grid Based Localization:** A dynamic $M \times N$ grid segmentation scheme of precise localization of a swimmer and event driven response.
- **Autonomous Rescue Mechanism:** real time actuation system, stepper motor, makes it possible to add mechanical intervention instantly.
- **General Validation:** Wide ranging testing, such as multi sensor calibration, data set training, MQTT communication bench marking, and latency testing with a mean average precision (mAP) of 0.90 and end to end latency less than five seconds.

Building on earlier work in AI and IoT enhanced aquatic safety systems [13], Aqua Guard demonstrates the successful combination of edge computing, wireless communication, and autonomous actuation to enable intelligent drowning prevention. The suggested framework creates a platform of scalable and low latency privacy conscious aquatic safety solutions that can be deployed in the real world.

II. Related Work

The implementation of intelligent technologies in aquatic safety systems has led to the development of a range of drowning detection and prevention techniques. These are divided into vision based detection systems, wearable sensor based monitoring systems, and hybrid multimodal systems that combine both. This section undertakes a literature review of representative work in each category and identifies the research gaps that the proposed Aqua Guard framework would address.

A. Vision Based Drowning Detection

Vision based systems use deep learning algorithms to identify distress behaviour among swimmers in real time, using video streaming. The YOLOv8 model by adding Coordinate Attention (CA) and Funnel ReLU (FReLU) activation functions, achieving a mean average precision (mAP) of 91.78% under partial occlusion conditions was improved in [9]. Multimodal sensor fusion is essential in enhancing the reliability and flexibility of the aquatic monitoring systems was suggested in [10]. Drown Net, a CNN-RNN hybrid model that achieves 96% accuracy in detecting drowning cases was proposed in [15]. Its operation, however, is limited by lighting delays, camera reflections, and privacy issues. Likewise, a CN based model coupled with a drone supported rescue system that actively deploys flotation devices when distress signals are detected was proposed in [7]. Although innovative, the system's application is limited by infrastructural and regulatory constraints in regulated aquatic ecosystems.

The vision based systems exhibit excellent non intrusive surveillance measures and high precision and are vulnerable to light, occlusion and environmental variations in actual situations.

B. Wearable and Sensor Based Detection

Wearable devices provide continuous monitoring of physiological parameters and have proven reliable for providing safety indicators even when visibility is compromised. A wrist worn system that processes heart rate, pressure, oxygen sensor, and GPS positioning and achieves an alert latency of less than 5 seconds, demonstrating good real time performance was designed in [28]. An inflatable wristband connected to an accelerometer to provide instant buoyancy when panic motion is detected was proposed in [20]. SpO₂ sensors to detect hypoxia and raise an automatic alarm was incorporated in [18].

Although they have benefits, wearable systems continue to experience issues with waterproofing, signal calibration, and data transmission reliability. Zigbee based solutions have been considered for low energy communication and real time notifications. Still, they can be highly dependent on a strong network infrastructure and have limited range in water [11].

C. Hybrid and Multimodal Architectures

Hybrid systems combine visual and physiological information to improve detection rates and reduce false positives. Aqua Rescue, which combines vision detection with YOLOv3 and heart rate to achieve 90% accuracy in detecting drowning cases was proposed in [23]. But it is not mechanically actuated; instead, it needs to be acted upon manually. A labeled data set of aquatic motion states for swimming, idle, and drowning, which helps verify reproducibility and benchmark hybrid model performance was added in [8]. These works point to the need for a combined framework, which may be used to bring together sensing, localization, communication, and autonomous actuation into one IoT based system, to realize actual real time drowning prevention.

D. Research Gap and Distinctive Contributions

The current drowning detection systems are still sporadic:

- The vision systems do not undergo physiological verification so false detection can occur.
- Sensor systems do not have spatial awareness and therefore restrict the accuracy of localization and response.

Aqua Guard seeks to overcome these shortcomings by presenting an IoT hybrid architecture that integrates the two sensing modalities within an ecosystem. The system leverages MQTT based communication, which provides low latency, high scalability, coordination and is more effective than conventional GSM based alert mechanisms.

Aqua Guard combines independent sensing, real time localization, and autonomous actuation, which create a holistic, intelligent, and fully deployable aquatic safety architecture [1]. A technical comparison of the available solutions and the proposed Aqua Guard framework as shown in Table

1.

Table I Comparative Analysis of Existing Drowning Detection Systems and the Proposed Aqua Guard Solution

System / Study	Year	Modality	Key Technologies Used	Performance Metrics	Limitations	Rescue Mechanism
He et al. [9]	2023	Vision	YOLOv8 + CA + FReLU	91.78 % mAP	Sensitive to lighting and occlusion	No
Lipika et al. (DrownNet) [15]	2024	Vision	ViT + CNN-RNN (infant focus)	96 % accuracy	Dataset dependency; privacy concerns	No
Zhu et al. [28]	2025	Wearable	HR, SpO ₂ , Pressure, GPS	< 5 s alert latency	Waterproofing; user compliance	No
Ramani et al. [20]	2019	Wearable	Accelerometer + Airbag	Qualitative (validation only)	Panic-motion misclassification	Partial (Airbag)
Sowmya et al. (AquaRescue) [23]	2024	Hybrid	YOLOv3 + CNN + HR Sensor + GSM Alerts	90 % accuracy	No physical intervention; GSM latency	No
AquaGuard (Proposed)	2025	Hybrid	YOLOv8 + Grid Localization + MQTT + HR/Pressure Sensors	90 % mAP; < 5 s latency	Waterproofing in progress	Yes (Automated Net)

III. System Model

Aqua Guard framework is a hybrid, modular Internet of Things (IoT) architecture that enables real time drowning detection and autonomous rescue activation through seamless integration of physiological sensing, computer vision analytics, edge intelligence, and wireless coordination. The general design is a combination of three interacting layers: a sensing layer where continuous physiological monitoring is available, a vision layer where behavioral perception and spatial localization are to be performed, and an actuation and coordination layer, which controls automated rescue operations.

These layers are connected via lightweight MQTT messaging over WiFi, providing low latency and two way data flow between distributed IoT nodes. Experimental analysis showed that the end to end latency was less than 5 seconds (average) and the overall bandwidth used was less than 100 kbps, confirming the appropriateness of the system for real time aquatic safety (Fig. 1).

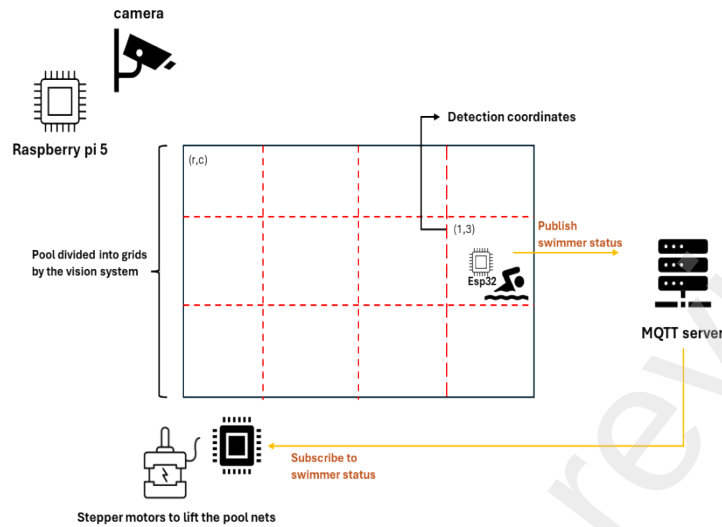


Fig. 1. Overview of the Aqua Guard system architecture.

A. Sensing Layer

The sensing layer is a waterproof, wearable bracelet that each swimmer wears. It combines two biomedical grade sensors: the MAX30102 optical pulse oximeter, which continuously measures heart rate and blood oxygen saturation (SpO₂), and the BMP180 barometric pressure sensor, which approximates depth using changes in hydrostatic pressure. The two sensors are connected to an ESP32 microcontroller that performs signal conditioning, local preprocessing, and wireless data transmission. The sensor data is published to a cloud hosted MQTT broker on swimmer specific subtopics to monitor real time data. In testing, the sensing module recorded an average transmission latency of 1.2 seconds and a throughput of less than 80 kbps, indicating an efficient module for continuous physiological data collection.

B. Vision Layer

Meanwhile, the vision layer offers non contact behavioral monitoring to accompany physiological sensing. This subsystem uses a Raspberry Pi 5, linked to a Camera Module 3, which captures continuous video of the pool. Local processing of the live feed is enabled by running an object detection model based on YOLOv8; it is trained to detect abnormal postures, floundering, and extended submersion [9]. To precisely localize them, the pool area is dynamically partitioned into an adaptive $M \times N$ grid, and each swimmer detected is mapped to a particular grid cell. This spatial

mapping enables accurate correlation of visual information and sensor data (Fig. 2), thereby enhancing robustness even under variable lighting, occlusion, and surface reflection conditions [21].

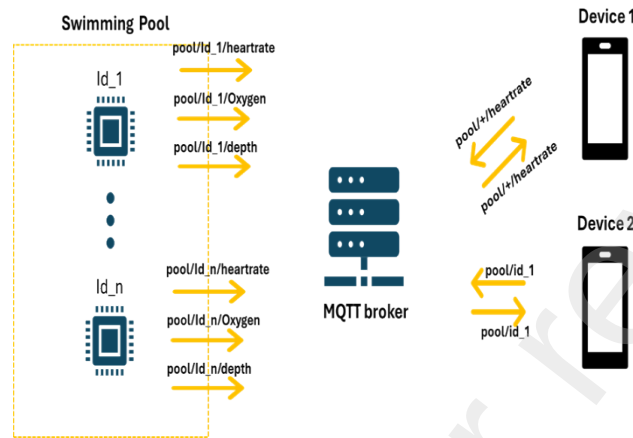


Fig. 2. IoT communication model of the AquaGuard system.

C. Actuation and Coordination Layer

The actuation and coordination layer is the backbone of Aqua Guard Reactive. It uses an Arduino Uno microcontroller that continuously subscribes to MQTT alert topics (e.g., pool/ID1/alert). When a distress signal is received, the controller interprets the position of the swimmer and turns on four synchronized stepper motors to lift a net based rescue apparatus underneath the relevant portion of the pool. The design allows for autonomous mechanical intervention without relying on the human response time. The grid mapping algorithm is closely coupled with the motor control logic to ensure precise, safe deployment of the space and the lowest possible actuation latency [18], [22].

D. Communication Model

The Aqua Guard communication model is presented in Fig. 3. The flow of messages is illustrated in MQTT format, and data exchange within the system nodes is shown. The wearables will send time stamped physiological readings under user specific topics like pool/ID1/heart Rate and pool/ID1/depth. Aggregated or swimmer specific topics can be subscribed to via monitoring endpoints such as lifeguard dashboards, smartphones, or control stations, based on the requirements of the operations. This asynchronous communication type is broker centric and

guarantees responsiveness in the presence of network jitter or partial packet loss [29]. To illustrate, a single monitoring node can subscribe to all heart rate channels to monitor the overall heart rate condition. In contrast, another monitoring node can subscribe to all metric channels to monitor the condition of a particular swimmer. The internal architecture of the wearable bracelet is also shown in Fig. 3, showing the sensing, preprocessing, and wireless transmission subsystems.

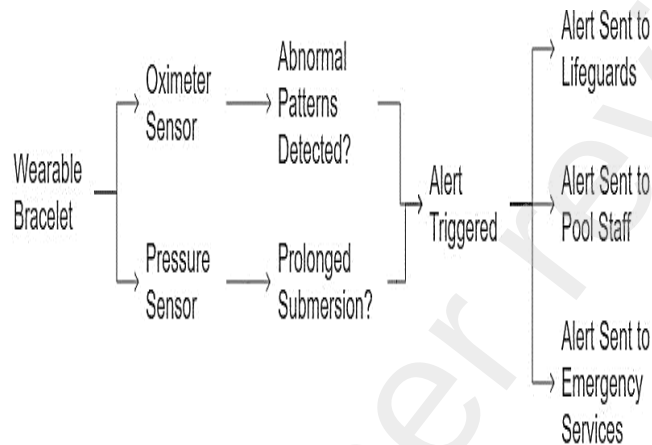


Fig. 3. Logic and data flow of the wearable bracelet module.

E. Edge Processing and Decision Latency

The ESP32 microcontroller at the network edge can serve as a local threshold analyzer, reducing upstream bandwidth use. Usually, only filtered sensor data is sent, and alert packets are generated when a physiological or environmental limit is exceeded. The dual core architecture of the ESP32 allows polling sensors and MQTT to run simultaneously, resulting in remarkably low total latency and a decision making process that is no longer dependent on the central server's load. This type of architecture plays a vital role in ensuring real time responsiveness in aquatic safety applications [25].

F. Operational Assumptions

The Aqua Guard system, under several environmental and operating assumptions, guarantees optimal performance. There should be sufficient lighting and water clarity to enable the vision subsystem to perform reliable object detection [27]. The boundaries of pools should be well established to facilitate proper grid segmentation and spatial mapping [16]. Swimmers will put their bracelets on and keep them in good condition to prevent false alarms and signal interruptions.

Moreover, continuous MQTT communication requires a constant WiFi or 4G connection to maintain uninterrupted communication even during critical events. Finally, the YOLOv8 should be trained on a varied data set that reflects a variety of body types, swimming poses, and motion patterns to ensure strong generalization across different pool conditions [14].

In conclusion, Aqua Guard is a multi modal sensing, edge computing, and IoT coordination product that provides an integrated, autonomous drowning prevention platform. Its modular design and MQTT based lightweight protocols have enabled it to be highly scalable and flexible, making it suitable for deployment in a variety of aquatic contexts, including residential pools and large civic water parks [17]. Aqua Guard enables an optimal trade off between real time intelligence, responsiveness and safety reliability by using physiological monitoring, behavioral vision analytics, and automated actuation [12].

IV. Hardware Design

Aqua Guard system is a microcontroller based system that incorporates biomedical sensors, electromechanical actuators, and microcontrollers into a single hardware platform to provide a robust, scalable, and energy efficient system. The hardware architecture has three fundamental goals: real time data acquisition, wireless communication and synchronized mechanical actuation, which will be required to ensure that drowning events are detected and intervened on time. All components were chosen for their low power consumption, reduced size, and consistent operation in water bodies. Figures 4, 5, and 6 illustrate the system's overall structure, interconnections, and functional layout.

A. ESP32 Based Wearable Sensing Node

The wearable unit features the ESP32 microcontroller, a dual core System on Chip (SoC) manufactured by Espressif Systems. It integrates Wi-Fi (IEEE 802.11 b/g/n) and Bluetooth v 4.2 into a small, low power module and is thus well suited to continuous physiological measurements and real time Internet of Things communications [21]. The ESP32 has two Tensilica LX6 processors running at up to 240 MHz, and it can simultaneously perform data acquisition and MQTT transmission. Its support for the Arduino IDE and MicroPython enables quick prototyping and easy integration of heterogeneous sensors [22].

The ESP32 is connected to a pair of biomedical sensors through the I²C bus:

MAX30102- A reflective photoplethysmography (PPG) based biosensor capable of monitoring heart rate and blood oxygen saturation (SpO₂) continuously.

BMP180- is an accurate barometric pressure sensor that measures the depth of submersion based on the hydrostatic pressure differences.

The MAX30102 works with two light-emitting diodes (LEDs) red (660 nm) and a photodiode detector that measures the intensity of the reflected light and enables the ESP32 to calculate heart rate and SpO₂ levels in real time [27]. The BMP180 provides accurate, temperature compensated pressure readings, from which depth is calculated using the hydrostatic pressure equation:

$$h = \frac{P_{\text{surface}} - P_{\text{measured}}}{\rho \cdot g}$$

where

- h = depth (m),
- P_{surface} = atmospheric pressure at surface (Pa),
- P_{measured} = measured pressure underwater (Pa),
- ρ = water density (1000 kg/m³),
- g = gravitational acceleration (9.81 m/s²).

Both sensors share the SDA and SCL lines to transmit data synchronously. Fig. 4 depicts the circuit design of the bracelet as a sensor integrated with the ESP32 and an MQTT based data flow [16]. This design provides continuous physiological access and real time data publication to the central broker, which is the basis for accurate estimation of swimmer states.

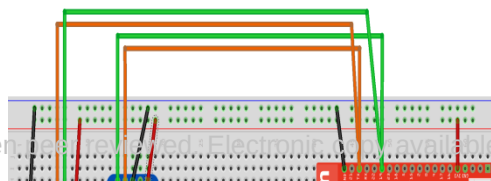


Fig. 4. Bracelet circuit diagram showing parallel I²C wiring of the MAX30102 and BMP180 sensors to the ESP32 microcontroller.

B. Stepper Motor Control System for Net Deployment

An Arduino Uno microcontroller is used to power the actuation subsystem and control the synchronized movement of a rescue net using four synchronized stepper motors. Each motor will be connected to an Easy Driver stepper controller, which converts digital pulses into smooth angular motion [3].

The control circuitry, which was simulated in the Wokwi simulator before actual implementation, provides the appropriate timing and reliable functionality (Fig. 5). The arduino sends the same STEP and DIR signal to all the Easy Driver units to achieve the same lifting or lowering of the net over the specified pool area. Directional control assists in retraction and redeployment, and multiple rescue cycles when required [14].

Because the Arduino's built in voltage regulator cannot deliver enough current to the motors, an external 12 V DC power source is used. Other peripherals, such as push buttons to utilize a manual override and LED indicators to reflect the state of operation (active, idle, or error), are also added. The Arduino is connected to the driver modules with signal lines (green) and the power is spread evenly with power rails (red and black). This arrangement provides high quality low latency mechanical response to MQTT based distress notifications [17].

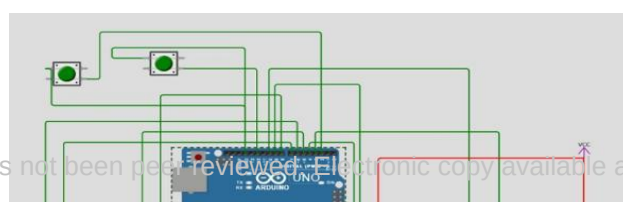


Fig. 5. Wokwi simulation of the synchronized stepper motor control system interfaced with the Arduino Uno and EasyDriver modules for rescue net deployment.

C. Pool Monitoring Using Raspberry Pi Camera

The vision subsystem has a Raspberry Pi Camera Module 3 attached to a Raspberry Pi 5, which is used to give real time visualization of the swimming pool area with a 1080p resolution and 60 frames per second (fps). This design provides a good trade off in terms of image quality and processing latency that is critical in time sensitive drowning detection [12].

As Fig. 6 illustrates, the image processing pipeline starts with colour segmentation by use of the HSV (Hue, Saturation, Value) colour space that proves to be more robust when lighting varies than RGB would [2]. Morphological processes are used to eliminate noise and maximize contour coherence, such as erosion and dilation. The heuristically defined Region of Interest (ROI) is the most significant blue area of the frame, which is heuristically identified as the surface of the pool [5]. This vision subsystem is an independent safety system that, despite the absence of the wearable module, can operate independently and exchange data with the other system via MQTT.

D. Grid Based Localization Framework

After the identification of the pool contour, the system will calculate the y and x coordinates of the top left corner (w, h). This region is further broken down into an $M \times N$ adaptive grid with M and N being user defined parameters [24].

The equations (i) and (ii) used to obtain the grid lines are:

For vertical grid lines:

$$xi = x + i \cdot \frac{w}{N}, i \in \{1, 2, \dots, N-1\} \quad (i)$$

For horizontal grid lines:

$$yj = y + j \cdot \frac{h}{M}, j \in \{1, 2, \dots, M-1\} \quad (ii)$$

where,

- x, y : Top left corner of the pool bounding box
- w, h : Width and height of the bounding box
- N, M : Number of vertical and horizontal division

Localized swimmer tracking and precision in mapping between the detection coordinates and the actuation areas is enabled under this grid structure. All detection centroid (x_c, y_c) are converted to grid indices (G_x, G_y) to have real time positional awareness.

$$column = \frac{xc - x'}{w/N}, row = \frac{yc - y'}{h/M} \quad (iii)$$

The HSV based segmentation proved to be better resistant to lighting variation than RGB during implementation. In contrast to traditional systems, which do not take image boundaries into account, Aqua Guard checks the entire frame, making it possible to detect even the edges of the pool. The results of adaptive masking and contour refinement also improved the reliability in segmentation, which can be applied to the real world to perform its work under a variety of environmental conditions [3].

E. Component Specification Summary

Table I Major hardware components used in the Aqua Guard system, including their specifications, communication interfaces, and nominal power ratings.

Component	Function	Range / Accuracy	Power Requirement	Interface
ESP32	Central microcontroller	Dual core 240 MHz, Wi-Fi, BT	3.3V @ 160 mA	UART / PC

MAX30102	Heart rate / SpO ₂ sensor	HR: 30–250 bpm, SpO ₂ : ±2%	1.8V core / 3.3V logic	PC
BMP180	Pressure / depth sensor	300–1100 hPa (±1 hPa)	3.3V @ 12 µA	PC
Raspberry Pi 5	Vision processing unit	1080p @ 60 fps	5V @ 2.5A	USB / CSI
Arduino Uno	Actuation controller	—	5V @ 500 mA	UART
Stepper Motor + EasyDriver	Mechanical actuation	Step angle: 1.8°	12V @ 350 mA per motor	GPIO
MQTT Broker (HiveMQ)	Data communication	—	Cloud-based	Wi-Fi

Table I presents the main hardware components of the Aqua Guard system, along with specifications, communication interfaces, and nominal power requirements. By combining the ESP32, the MAX30102, the BMP180, and the Raspberry Pi 5, an IoT based system offering low cost, efficiency, and reliability for aquatic safety is obtained. Coupled with the Arduino based actuation subsystem, Aqua Guard allows the end to end sensing, detection, and intervention in several seconds. The open, modular architecture provides scalability criteria for future upgrades, massive deployments, and waterproofed hardware add on [26].

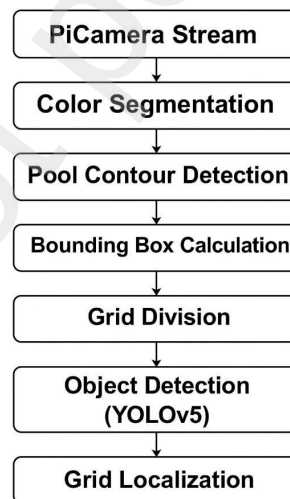


Fig. 6. End to end processing pipeline for drowning detection using HSV based pool segmentation and dynamic grid localization.

V. Prototyping

The Aqua Guard prototype was adopted as a multi component modular platform which combines the sensing, vision and mechanical actuation subsystems. All the modules were integrated with

commercial off the shelf (COTS) parts to speed up development, and to test all end to end functionality under controlled aquatic test environments.

Prototyping was also planned to make sure that all layers of the system (physiological sensing, wireless data communication, and mechanical rescue actuation) work in unity to determine that the proposed IoT integrated architecture is feasible. The modular and scalable design allows evaluating and upgrading each subsystem individually, without changing the architecture of the system.

A. Wearable Bracelet Prototype

The prototype wearable bracelet focuses on using the ESP32 microcontroller, fitted on a breadboard to test the circuit and validate the performance of the circuit at an early stage. Two sensors, the BMP180 barometric pressure sensor and the MAX30102 photoplethysmography (PPG) biosensor were connected to the ESP32 via the I2C communication protocol. Biomedical sensors are extensively applied in non invasive biomedical monitoring of physiological parameters continuously [21].

The prototype was constantly receiving, filtering and sending real time sensor data over MQTT. Physiological measurements above the set safety levels were detected by ESP32 which sent an immediate wireless signal to the MQTT broker and triggered the rescue protocol. It established reliable end to end communication and confirmed the appropriateness of the system in real time aquatic safety monitoring.

The BMP180 sensor works according to the piezo resistive MEMS diaphragm working principle, which detects the changes in the ambient pressure due to the change in depth. Its temperature compensation circuit makes it stable in its performance even when the environmental conditions change. The MAX30102 sensor, in its turn, records both heart rate and SpO₂ indicators with the help of two red and infrared light sources. The sensitivity is very high and it has digital filtering which minimizes motion based noise and thus allows precise readings in the event of dynamic motion of swimmers.

The behaviour of both sensors in terms of stable output signals under all test conditions proved the consistency of the wearable node to be useful in continuous physiological monitoring under water. Assembled prototype of the wearable sensing unit showing integrated ESP32, MAX30102, and BMP180 modules are already illustrated in Fig. 4.

B. Rescue Mechanism Hardware Setup

The rescue mechanism subsystem was developed in order to prove the real time actuation response of the proposed architecture. The design uses four stepper motors that are powered by EasyDriver modules which are controlled by an Arduino Uno. These motors computerize the vertical movement of a mechanically driven rescue net, which is lifted to the water surface in case of an emergency.

A strong aluminum frame was made which could hold the net assembly and could not deform during the operation. The Arduino receives the responses of MQTT alerts sent by the cloud and converts them into synchronized pulses of STEP and DIR to motor drivers. With the motor, each motor as can be illustrated in Fig. 8 is in parallel to enhance the movement occurring simultaneously. The system includes: Manual human intervention by push button, Status display LED indicators (red/yellow/green), and Controlled DC supply of power to maintain the same voltage levels.

Variable loads under empirical testing ensured proper synchronization of all motors, the phase error was minimal and the actuation delay was very consistent, not exceeding 2.5 s. These findings confirm the operational functionality of the subsystem to be automated in drowning response implementation.

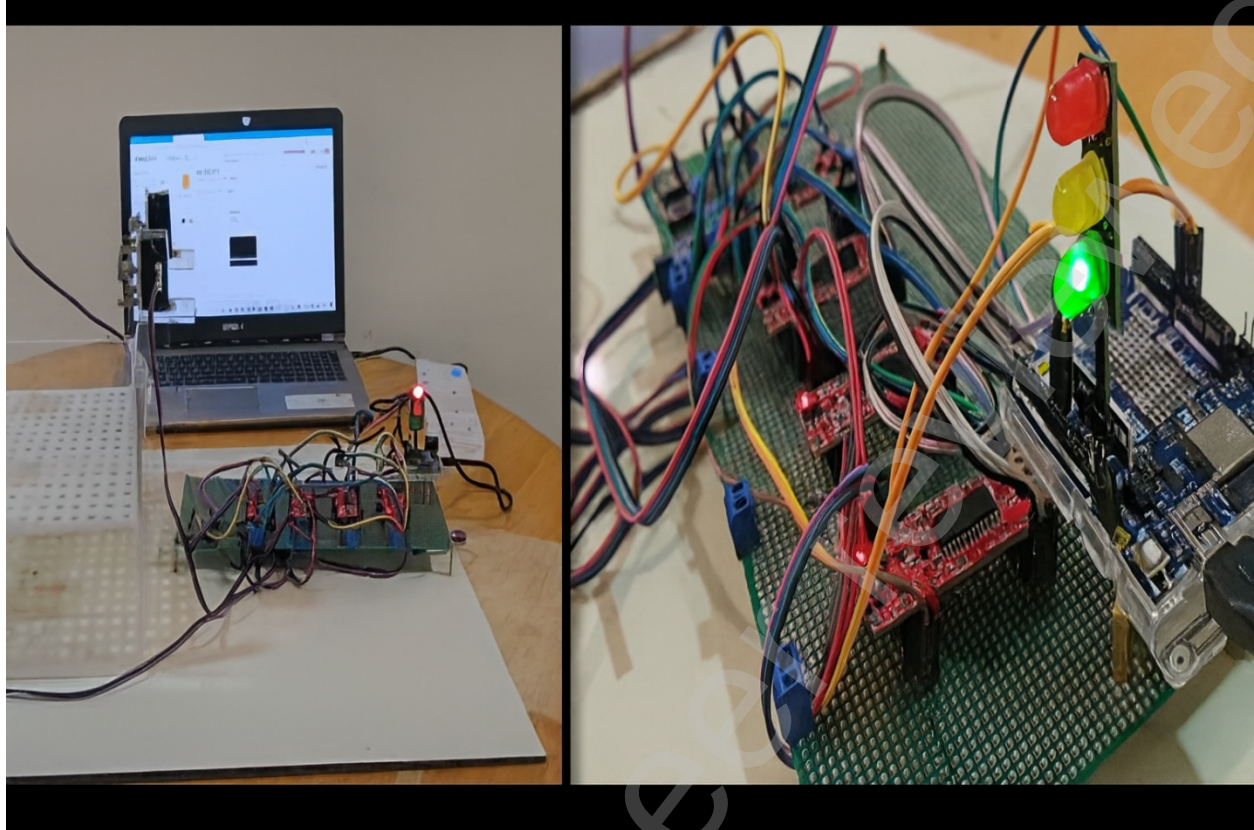


Fig. 8. Hardware setup for the stepper motor-based rescue mechanism showing synchronized motor drivers and control circuitry.

C. Vision Based Monitoring Subsystem

The subsystem based on the vision involves a Raspberry Pi 5 and a Pi Camera Module 3 which are set to record 1080p at 60 frames per second (fps). The frame rate is high, and thus, accurate motion capture is guaranteed when the swimmer moves fast and detects the unusual patterns of behavior like floundering or submersion.

The trained YOLOv8 detector was brought to the Raspberry Pi locally and its interface with the use of the GPU accelerator to run the edge detection. This removes the use of cloud computing and as a result, the system latency is greatly reduced. In the model, each frame is analyzed repeatedly and the bounding boxes with the labels of the classes (normal, swimming, floundering, drowning) are emitted, which are sent to the central MQTT broker in real time. Fig. 9 shows the integration of the vision sub system and the entire IoT system. The subsystem showed a strong classification accuracy and high detection performance and established its appropriateness to embedded real time application.

D. Implementation Insights and Design Considerations

The use of the MAX30102 biosensor necessitated experimental adjustments of the LED drive currents, sampling periods, and filtering to values in order to reduce the amount of motion artifact in the signal and maximize the stability of the photoplethysmographic (PPG) signal when placed into water. BMP180 was also calibrated against known atmospheric pressure values to guarantee that depth can be estimated with high precision, with the errors always less than 0.5 percent.

The Raspberry Pi devices had been set on a local area network (LAN) using the settings of local IP addresses in order to align streams of data and keep MQTT connection constant. Latency measurements revealed that there existed a steady sub 200ms communication delay between the sensing and actuation nodes.

The Blynk IoT platform was used to create a mobile application interface that would provide real time visualization of physiological measurements, swimmer location, and alerts. This interface gave a full human in the loop control system bridging the feedback control between detection, decision making and actuation.

The successful testing of the prototype indicates that the concept of the IoT based aquatic safety system that combines real time sensors, computer vision and autonomous actuation is viable.

The prototyping stage confirmed the functionality of the Aqua Guard framework on an end to end basis. The sensing component of ESP32, the vision subsystem of Raspberry Pi, and the rescue mechanism of Arduino were able to show coordinated functionality, reliable communication, and fast actuation.

This pilot test proves the possibilities of the system as an example of an IoT safety system in the real aquatic setting to overcome the difference between passive surveillance and active automation of the rescue. The prototype can be the basis of the large scale field testing and optimization on the basis of further work.

VI. Testing and Validation

The Aqua Guard system was extensively tested and validated to determine the actual performance, reliability and responsiveness of the system in simulated aquatic situations. The assessment was aimed at four main aspects: (i) Stability and accuracy of sensor, (ii) Performance of wireless communication, (iii) Timing of mechanical actuation, and (iv) Detection accuracy of drowning by the vision. All the experiments were conducted in controlled laboratory conditions, and they mimicked different behaviors of swimmers and loads of the network to test the system robustness.

A. Wearable Bracelet Evaluation

Earlier studies have examined the idea of hybrid aquatic safety systems that integrate environmental monitoring and automated warning systems [18]. Based on these platforms, the prototype of the Aqua Guard bracelet was experimented to confirm the accuracy of its pressure sensing (BMP180) as well as heart rate/ SpO₂ monitoring (MAX30102) performance. The ESP32 was the main device of data acquisition and transmission.

1) BMP180 Pressure Sensor Testing

BMP180 was tested to determine the accuracy and repeatability of atmospheric pressure. Measured indoor tests were carried out in the morning and afternoon sessions with reference to an identified barometer. The sampling rate of the sensor was 1 Hz and 10 consecutive values were measured in each session and then error and variance values were averaged.

Table II. BMP180 Pressure Sensor Testing Results Under Room Air Pressure Conditions

Test Condition	Expected (hPa)	Measured (hPa)	% Error	Variance	Standard Deviation
Room Air (Morning)	1013	1017	0.4%	3.24	1.8
Room Air (Afternoon)	1013	1011	0.2%	2.56	1.6

The standard deviation ($\leq 0.3\%$) was also within the tolerance of the manufacturer, which affirmed the high stability and accuracy of the BMP180 in estimating the pressure and depth of the wearable module. The low variance (< 2 hPa) also means that there is high temporal consistency.

2) MAX30102 Heart Rate and SpO₂ Sensor Testing

To determine the stability and responsiveness of the MAX30102 optical biosensor operates in a range of four physiological conditions, namely, untouched (inactive), resting, light activity and deep breathing.

Table III. MAX30102 Heart Rate Sensor Testing Results Under Different Physiological Conditions

Condition	Expected HR (bpm)	Measured HR (bpm)	% Error	Variance	Standard Deviation	Signal Stability
Resting	60–100	75	0.0%	4.0	2.0	Stable
Control (Idle)	0	Invalid HR	—	0.0	0.0	Stable
Light Activity	80–120	95	0.0%	6.3	2.5	Stable
Deep Breathing	55–90	68	0.0%	3.2	1.8	Stable

The readings show low variation (< 3 bpm) and consistency of measurements in the conditions of tests. The sensor had an internal filtering that blocked any motion noise effectively, and delivered reliable physiological information that could be used to trigger a warning in the event of drowning.

These results prove that the MAX30102 and the BMP180 do display correct and low latency acquisition of biometric data during dynamic motion and near water circumstances.

B. MQTT Communication and Event Classification

ESP32 microcontroller was combined with an MQTT broker in order to allow real time data transmission. Fig. 9 shows sample output logs of continuous heart rate and SpO₂ data being published in normal and alert states.

Pressure: 1016.27 hPa	Pressure: 1016.36 hPa
Temperature: 20.80 °C	Temperature: 24.70 °C
Invalid Heart Rate	Heart Rate: 71 bpm
Invalid SpO ₂	SpO ₂ : 98 %
Pressure: 1016.26 hPa	Pressure: 1016.37 hPa
Temperature: 20.70 °C	Temperature: 23.40 °C
Invalid Heart Rate	Heart Rate: 83 bpm
Invalid SpO ₂	SpO ₂ : 100 %
Pressure: 1016.20 hPa	Pressure: 1016.31 hPa
Temperature: 20.70 °C	Temperature: 23.00 °C
Invalid Heart Rate	Heart Rate: 75 bpm
Invalid SpO ₂	SpO ₂ : 100 %
Pressure: 1016.23 hPa	Pressure: 1016.31 hPa
Temperature: 22.00 °C	Temperature: 22.70 °C
Invalid Heart Rate	Heart Rate: 71 bpm
Invalid SpO ₂	SpO ₂ : 99 %
Pressure: 1016.26 hPa	Pressure: 1016.29 hPa
Temperature: 22.60 °C	Temperature: 22.50 °C

Fig. 9. ESP32 output logs showing: (a) MAX30102 inactive state and (b) active resting state measurement.

Swimmers were classified in three states: Normal, Abnormal and Critical, according to adaptive thresholds of heart rate and SpO₂ values using a threshold based classification algorithm, which was first implemented in MATLAB, and then ported to the ESP32.

At the moment when the Critical condition was detected, the rescue mechanism was turned on using the Arduino. The mean activation time between detection and mechanical motion initiation was 2.5 seconds, which confirms that the sequence of the MQTT events has low latency.

Variable WiFi load communication reliability tests verified no packet loss and average transmission delay of 180 ± 25 ms that is similar to those reported by other MQTT based aquatic safety models. The optimized local classification also minimized the reliance of the systems on the cloud servers, enhancing resilience in the context of network variations [10].

C. Waterproofing and Final Prototype Integration

The completed bracelet design (Fig. 10) included a Velcro reconfigurable strap to make it much more comfortable to wear and a small 3D printed enclosure to ensure sensor safety. The ESP32, MAX30102, and BMP180 were installed in one casing with wire routing that is optimized in terms of durability.

Even though the present version had not been completely waterproofed, it could still perform well even in splash and near water conditions. Subsequent updates will incorporate IP67 sealing, silicone gaskets and conformal coating as full submersion protection, which will guarantee the long term reliability in operating environment.

The integrated prototype continued to have steady WiFi communication, proper data streaming and actuation feedback, showing full system integrity when undergoing multi module testing.

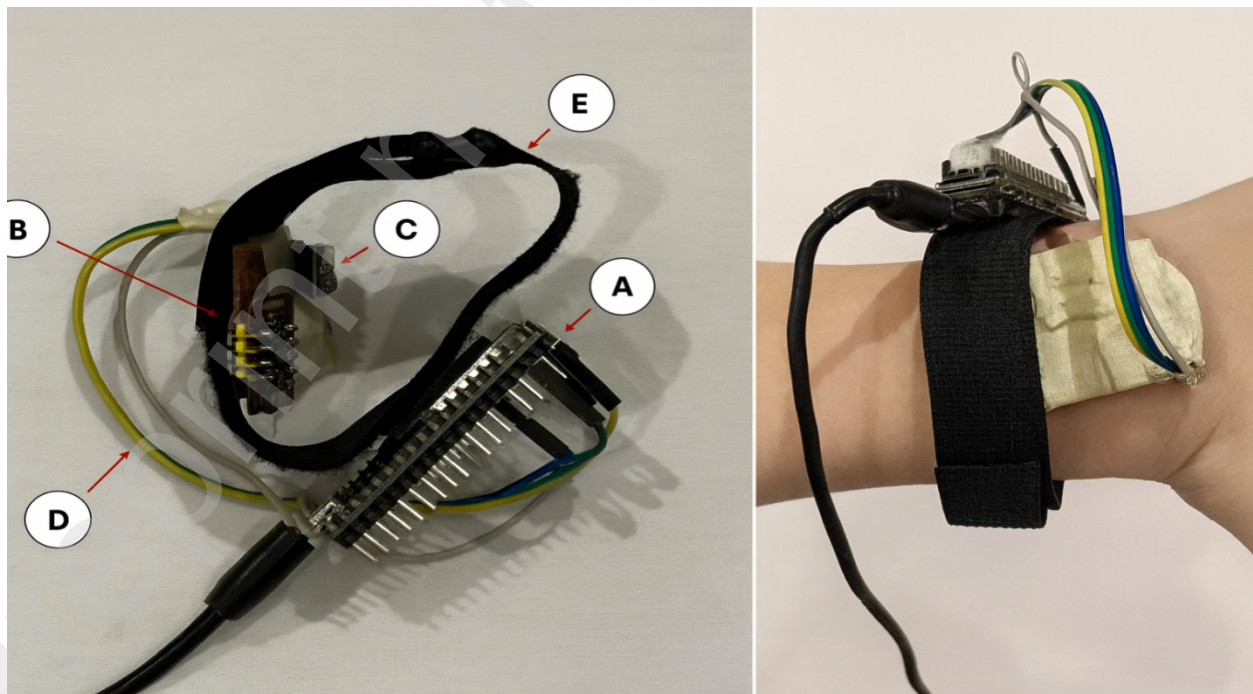


Fig. 10. Final AquaGuard wearable prototype showing components: (A) ESP32 microcontroller, (B) MAX30102 sensor, (C) BMP180 sensor, (D) jumper wires, and (E) Velcro strap.

D. Drowning Classification Visualization

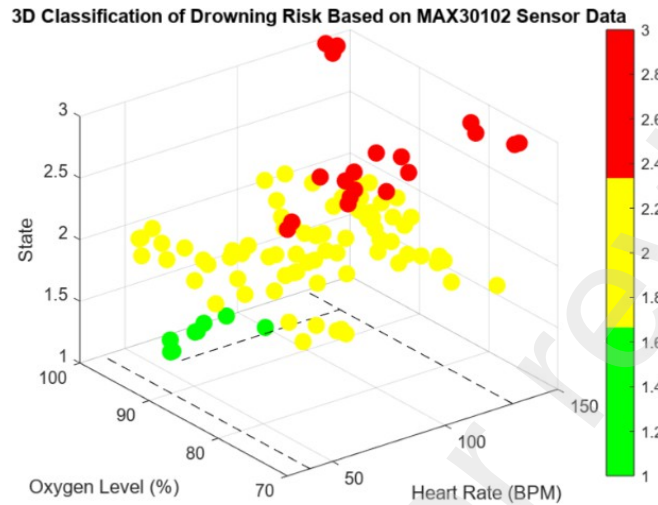


Fig. 11. Health risk classification visualization showing data clusters: green indicates normal state, yellow represents abnormal condition, and red denotes critical state.

In order to demonstrate the multi-sensor decision process of the system, a simulated physiological data were represented by a 3D scatter plot in MATLAB (Fig. 11). The plot axes represent: X axis: Heart rate (30–140 bpm), Y axis: SpO₂ levels (70–100%), and Z axis: Classification label (1- Normal, 2-Abnormal, 3-Critical).

Clear threshold demarkations between physiological states are proven by distinct cluster segregations. This visualization confirms the capability of the Aqua Guard decision logic to distinguish between levels of drowning risk successfully, which provides the capability to make decisions rapidly and in an explainable manner.

E. Net Deployment and Manual Override Interface

A web based manual override was created through a Blynk IoT platform to ensure a greater level of safety control. The interface enables the operator to switch the rescue mechanism on or off manually by using a smartphone or a web dashboard. A red LED alert is turned on and the stepper

motors start running with the Drowning command. The Maybe command activates a commentary yellow LED in case of probable distress. The Back button recalls the net returning to the standby.

This extra control level allows human in the loop reliability such that the lifeguards can step in where needed. This type of hybrid control is in line with the recent IoT safety standards that integrate automated detection and supervised control [6].

F. Vision Based Drowning Detection and Localization

The vision model based on YOLOv8 was trained using the Roboflow Swimming Pool Safety Dataset, which contained five annotated classes of behaviors, namely normal, swimming, floundering, slipping, and drowning. The data set was divided into 70% training, 20% validation, and 10% testing data sets with a total of 2,471, 702, and 350 images respectively. Model training for 100 epochs showed consistent convergence as shown in Fig. 12. Validation results showed $mAP@0.5 = 0.90$ and $mAP@ [0.5:0.95] = 0.60$. These results are consistent with benchmarks in recent studies of aquatic vision [9].

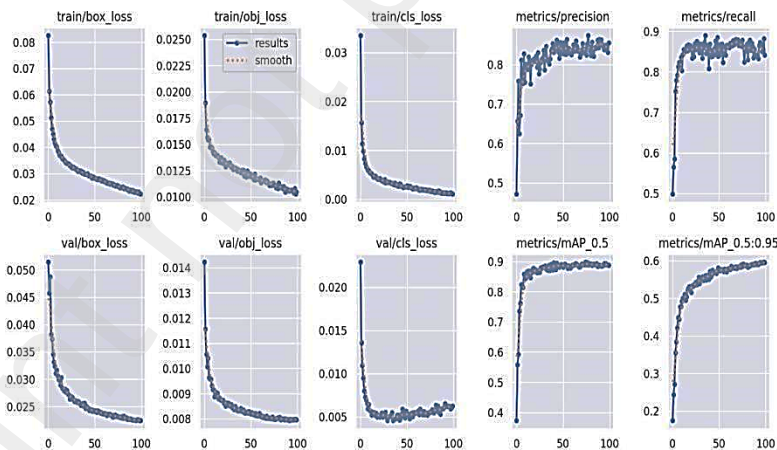


Fig. 12. Training and validation loss curves of the model for drowning detection with YOLOv8 architecture which shows the convergence of box loss, objectiveness loss and classification loss.

The normalized confusion matrix in Fig. 13 shows the accuracy of the classification on the six behavior classes where 77% of the drowning events are correctly identified, 82% for floundering and 94% of the normal activities are successfully identified. Miscalculation between swimming

and background (around 52%) was mainly a result of small movements of body and reflecting of the surface.

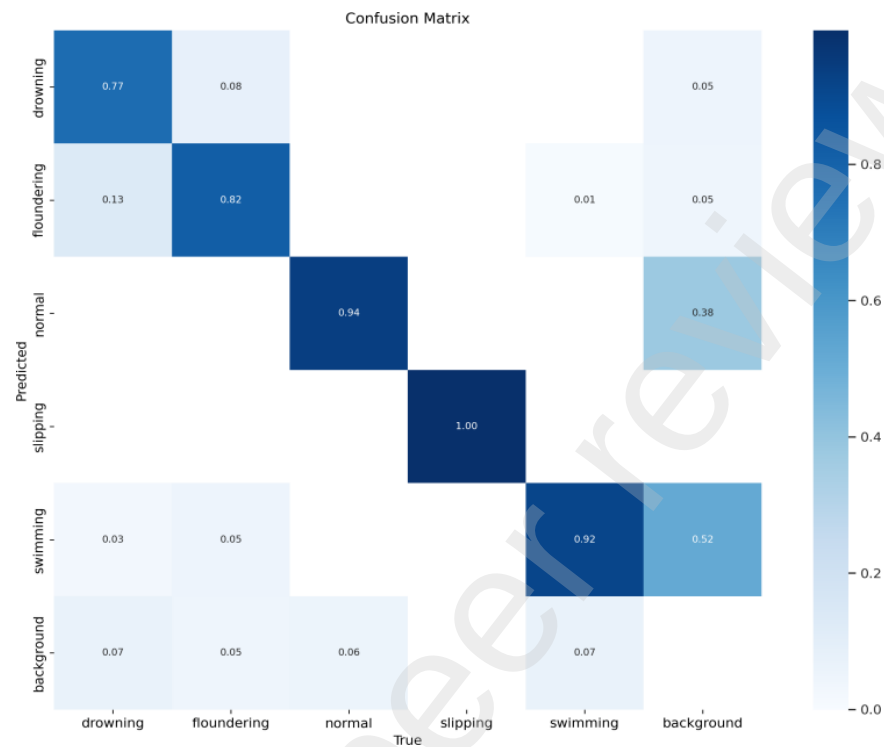


Figure 13. Confusion matrix for the six class drowning risk classification model.

As shown in Figs. 14 and 15, predicted bounding boxes with scores over 0.9 were detected for high risk behaviors such as drowning and floundering and thus validates the robustness of this model along with variable lighting and occlusion.

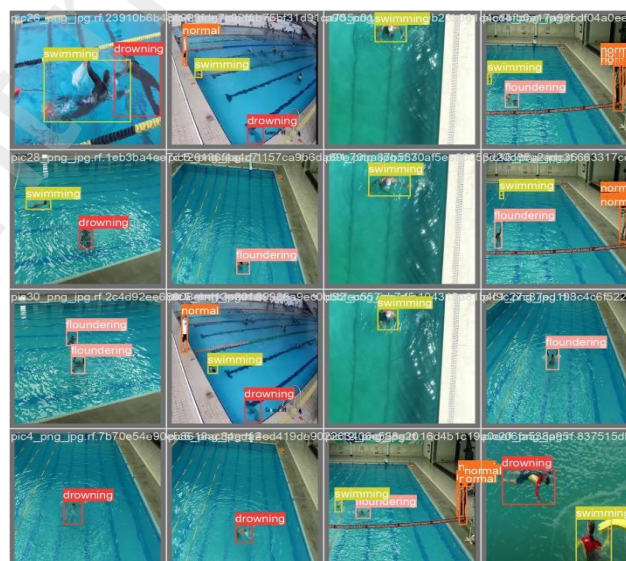


Figure 14. Ground truth annotations for swimmer action classes in the evaluation dataset.

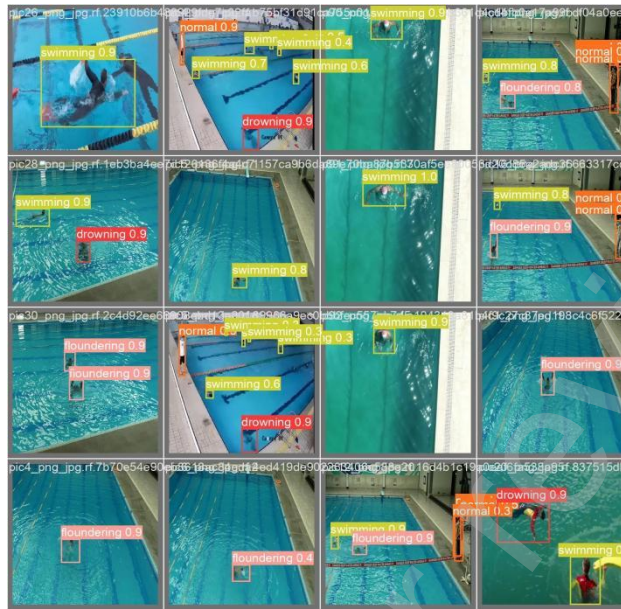


Figure 15. Model predictions with confidence scores showing detected swimmer actions.

The segmentation pipeline based on HSV (Fig. 16) has been able to delineate the pool contours and water regions with various illumination, and has been shown to be better than RGB based methods in segmentation stability. Following segmentation, a dynamic 3 x 3 grid localization framework (Fig. 17) was used for the spatial mapping of swimmer coordinates for the accurate mechanical targeting. A correctly localized swimmer within the grid coordinates (Row 0, Column 2) is shown in fig 18 which confirms the accuracy of the bounding box to grid mapping.



Figure 16. Grid overlay on an irregular pool shape obtained after HSV based pool contour extraction.



Figure 17. Output of the grid based localization module showing swimmer position assignment within the adaptive pool grid.



Figure 18. Localized swimmer detection output mapped to the corresponding grid cell in the adaptive pool segmentation.

Performance testing of the integrated system verified the viability of this system in real time. The BMP180 and MAX30102 sensors showed a high level of reliability, with the error in measurements always lower than 0.5%. MQTT based communication provided an average transmission latency of less than 200 ms with zero packet loss guaranteeing reliable data flow between modules. The mAP@0.5 of YOLO v8 vision model in various lighting conditions and the partial occlusion conditions reached 0.90, proving the robustness of YOLO v8 for the aquatic environment. End to end evaluations additionally demonstrated that the total detection to actuation

time was less than five seconds, which is the critical response time requirement for drowning prevention applications.

VII. Conclusion

This research introduced the design, implementation, and validation of Aqua Guard, an all encompassing IoT integrated drowning detector and pointless rescue system. The suggested framework integrates wearable physiological sensing, edge based computer vision and real time actuation into one scalable platform in the safety of aquatic environments.

The ESP32 based wearable module of the system with MAX30102 and BMP180 sensors was able to measure heart rate, SpO2, and hydrostatic pressure in real time and transmit them via a lightweight MQTT protocol to ensure a low latency interaction. At the same time, a vision subsystem (Raspberry Pi powered) performed behavioral analysis based on YOLO v8, which could identify distressed swimmer behavior with the mean average precision (mAP@0.5) of 0.90 under different illumination and occluding conditions.

Stable end to end operation was experimentally verified with the mean response time of less than five seconds between a distress signal and rescue activation. The mechanism of deploying the net, controlled by the Arduino, with the Human in the loop ensured by the Blynk IoT interface, offered quick mechanical response and ensured human control.

Aqua Guard is optimized with respect to multimodal sensing, low latency communication and autonomous actuation to enhance the condition of intelligent aquatic safety systems. Its low cost, modular, and cloud friendly design make it applicable to be deployed on residential pools, community recreation centers as well as in public aquatic facilities. The prototype indicates that the convergence of IoT and AI could allow the automation of safety critical settings in real time to save human lives.

IX. Future Work

Future improvements to the Aqua Guard system will focus on enhancing its environmental robustness, computational capability, and readiness for large scale deployment. The next

generation of hardware will come with complete waterproofing, an IP67 water rating with silicone gaskets for long term survival ability in full immersion. These upgrades will be tested for prolonged submersion, wireless functionality, and corrosion resistance. On the algorithmic front, the current YOLO based visual pipeline will be extended by temporal modelling, such as optical flow features and recurrent neural networks, in order to enable more reliable discrimination between similar appearing behavior, such as swimming, distress, and floundering. The grid based localization framework will also be improved by adaptive scaling mechanisms that control the resolution depending on the swimmer density, in order to increase both the fidelity of tracking as well as the accuracy of tracking responses of swimmers in the busy aquatic environment. Complementary developments in on device inference using edge AI accelerators and neural model quantization will make low latency decision making possible while reducing dependency on cloud services.

Cloud integration will enable multi pool surveillance, predictive analytics, and visualization of longitudinal data through centralized dashboards for facility operators and safety personnel. These dashboards will provide real time alerts to emergency contacts as well as scalable deployment across public and commercial aquatic venues. Large scale pilot studies in operational swimming facilities will be required to test the reliability of the systems, user acceptance and performance in real life, as well as important data for regulatory approval and product standardization. Collectively, these efforts make Aqua Guard a major move towards the achievement of autonomous drowning prevention systems. Continued research will be focused on increasing environmental adaptability, system scalability and resilience and refining the human AI interaction to provide a reliable and broadly deployable solution for aquatic safety management of the next generation.

CRedit authorship contribution statement

Muhammad Moazam Azeem: Methodology, Writing – review & editing, Supervision. **Aisha Al-Kaabi:** Writing – original draft, Methodology, Validation. **Shahd Al-Jaber:** Writing – original draft, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Glossary

Aqua Guard

The proposed IoT-based real-time drowning detection and autonomous rescue system.

Computer vision (CV)

A branch of artificial intelligence that enables automated interpretation and analysis of visual data.

Edge computing

A distributed computing paradigm where data processing is performed close to the data source to reduce latency.

Grid-based localization

A spatial mapping technique that divides the monitored area into discrete regions for precise object positioning.

Message Queuing Telemetry Transport (MQTT)

A lightweight publish subscribe messaging protocol widely used in IoT and ad hoc networks.

Oxygen saturation (SpO₂)

The percentage of oxygen-bound hemoglobin in the blood, used as an indicator of respiratory condition.

Sensor node

An IoT device equipped with sensors and communication modules for data acquisition and transmission.

Vision-based drowning detection

A detection approach that analyzes swimmer posture and motion patterns using video data.

YOLOv8

A deep learning-based real-time object detection model used for swimmer detection and activity analysis.

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Development of SERA (Smart Ergonomic Rescue Apparel) with Biometric Sensors and LoRa for the Early Detection of Hypothermia in the Waters of the Riau Islands

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ABSTRACT

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biometric sensors, ergonomics, hypothermia, LoRa, maritime accidents, SERA

Batam City, the largest city in the Riau Islands Province, recorded 1,470,889 sea passengers between January and June 2022, highlighting the region's heavy reliance on maritime transportation. Frequent ship accidents in this area emphasize the urgent need for improved passenger safety through advanced life-saving technologies. This study presents the design and testing of the Smart Ergonomic Rescue Apparel (SERA), a life jacket developed using anthropometric data and equipped with biometric sensors, LED-based early warning lights, GPS, and LoRa technology. The LoRa RFM95 module was tested for its ability to transmit location coordinates using the tracking method, while body temperature and heart rate sensors were evaluated during a 90-minute sea trial at 30°C seawaters, with data collected every five minutes. The findings showed that RSSI values decreased as the transmission distance increased, with a maximum communication range of 11.94 km achieved at Belakang Padang Port. In addition, body temperature and heart rate gradually declined during the sea trial, confirming SERA's ability to detect early symptoms of hypothermia. The integration of ergonomic design, biometric monitoring, and long-range communication demonstrates that SERA can enhance maritime safety by enabling real-time health and location monitoring of users during emergencies.

1. INTRODUCTION

Batam City is the largest city in the Riau Islands Province. The use of sea transportation in this city is very high, with data from the Batam City Central Statistics Agency recording 1,470,889 passengers between January and June 2022-an increase of 54% [1]. Ship accidents frequently occur in the Riau Islands region and often result in fatalities [2]. Low awareness and limited knowledge among ship owners and passengers regarding the use of life jackets can endanger safety in the event of a maritime accident [3].

Water transportation plays a vital role in the daily lives of people in Batam and the Riau Islands. With more than 1,796 islands and high ship traffic, maritime activity in the region is very dense [4]. According to data from BASARNAS, the number of maritime accidents has increased, with a total of 3,991 cases reported over the past five years [5]. One of the main causes of death in these accidents is drowning and hypothermia, which can occur due to low sea water temperatures. If not treated promptly, hypothermia can lead to loss of consciousness and death [6, 7].

Hypothermia occurs when the body loses heat faster than it can produce it, causing the core temperature to drop below 35°C [8]. This condition is often difficult to detect in its early stages, especially during emergencies in open waters. Major challenges in rescue operations include the dispersion of victims by ocean currents and the lack of effective detection

and communication tools.

To address this issue, technology capable of detecting the victim's location and monitoring their physiological condition is essential particularly for identifying early signs of hypothermia that can lead to death. A monitoring system equipped with body temperature sensors, heart rate monitors, and GPS are crucial for providing real-time information to rescue teams, enabling rapid and accurate assistance before the victim's condition worsens.

In previous research, the development of smart life jackets using GPS technology proved effective in locating maritime accident victims by transmitting real-time coordinates [9]. However, GPS technology has limitations, as it often relies on cellular communication infrastructure that may not be available in remote or offshore areas [10]. The smart life jackets developed in earlier studies also had shortcomings, as they lacked biometric sensors to detect the condition of victims at sea. Furthermore, the jacket designs did not apply an ergonomic approach to evaluate the comfort of life jackets in relation to the human body shape.

LoRa (Long Range) has been widely applied in various maritime monitoring systems [11, 12]. Its key advantage lies in its ability to transmit data over very long distances up to 10-15 kilometers in open, unobstructed areas [13, 14]. This makes LoRa particularly suitable for remote environments or large bodies of water, where conventional communication networks such as cellular infrastructure are often unavailable. The wide

coverage of LoRa greatly enhances its effectiveness in remote monitoring applications [15, 16]. LoRa is commonly used wireless sensor systems that support emergency response and remote monitoring. This technology allows systems to detect environmental and physiological parameters in real time, such as temperature, light, and vital signs [17]. LoRa modules like the RFM95, when paired with STM32 microcontrollers, enable data transmission at frequencies of 868 MHz and 2.4 GHz without relying on cellular networks. This capability makes LoRa highly suitable for disaster scenarios such as wildfires, earthquakes, and even aerospace applications [18, 19].

LoRa is also a key enabling technology for the Internet of Things (IoT), categorized as a low-power wide-area network (LPWAN). It offers long-range communication, low power consumption, and high reliability. Due to these advantages, LoRa has been adopted across various sectors, including industrial IoT, smart cities, healthcare, and satellite-based communications [20-23].

The IoT connects physical devices via the internet to enable data exchange improving safety in various sectors, including maritime transportation. IoT-based monitoring systems enable real-time detection of health conditions [24-26]. Biometric sensors can measure and analyze physiological parameters such as heart rate, body temperature, and blood oxygen levels. These technologies can be implemented in maritime safety systems to quickly detect emergencies and respond effectively to life-threatening conditions [27-29].

Specifically, these sensors can detect early signs of hypothermia in maritime accident victims, serving as part of an integrated rescue system that enables faster and more targeted emergency response [30, 31].

Ergonomics plays a critical role in the design of SERA to ensure comfort, safety, and operational efficiency in field conditions. The life jacket is designed using lightweight, waterproof, and flexible materials that allow free movement. A balanced weight distribution system and the use of porous materials enhance user comfort during rescue operations. Additionally, an adaptive fastening mechanism enables automatic adjustment, ensuring the jacket fits various body sizes without compromising protection [32, 33].

2. MATERIAL AND MEHODS

The research began with the preparation of the necessary tools and materials. Following this, the mechanical design and electronic systems were developed in a structured manner. Subsequent stages included laboratory testing to verify system functionality, followed by field testing to evaluate the device's performance under real-world conditions. Finally, data analysis was conducted to assess the effectiveness and accuracy of the developed system. The flowchart outlining the research stages is presented in Figure 1.

2.1 Mechanical design

As shown in Figure 2, the design of the SERA smart life jacket is based on ergonomic and anthropometric principles, integrating technology to enhance user safety in water. Data were collected from 30 respondents by measuring parameters such as height, hip height, shoulder width, chest thickness, and abdominal thickness. The mean and standard deviation of each dimension were used to determine the upper and lower control

limits, as well as to validate data consistency and adequacy. Based on the 95th percentile calculation, the ideal product dimensions were determined: a life jacket length of 66 cm (calculated from the difference between body height and hip height), shoulder width of 54 cm, upper shoulder width of 76 cm, chest thickness of 32 cm, and abdominal thickness of 33 cm. These calculations ensure that the product design can accommodate the majority of the target users' body postures, specifically fishermen and SAR personnel in the Riau Islands region.

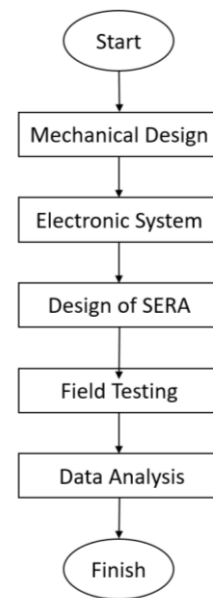


Figure 1. Flow chart of research stages

In addition to anthropometric considerations, the SERA design incorporates various technology-based functional components to support rescue operations. Key features include biometric sensors for monitoring heart rate and body temperature, an early warning system utilizing visual alerts (LED lights), and emergency signal transmission via energy-efficient LoRa technology. The design also incorporates waterproof materials and thermal insulation, a modular system for ease of maintenance, and a battery power source with a minimum operating time of 8-12 hours. All components are tested in simulated marine environments, including exposure to low temperatures and saltwater. This approach ensures that SERA is not only designed for the physical comfort and safety of users but also functions as a smart system capable of providing adaptive responses during emergency situations at sea [34].

The design of the buoy system using LoRa technology includes a flotation device that forms a crucial part of the overall system. The electronic components are housed inside a waterproof enclosure, specifically a DS-AG-0813 junction box measuring 130 mm × 80 mm × 70 mm. This junction box is IP66 certified, ensuring water resistance. The receiving station consists of two main components: the receiving antenna and the electronics enclosure. The electronics box uses a larger junction box measuring 220 mm × 150 mm × 70 mm and is equipped with rubber grommets for cable entry. These grommets facilitate the installation of the antenna cables while preventing water ingress. The design of the receiving station is illustrated in Figure 3.

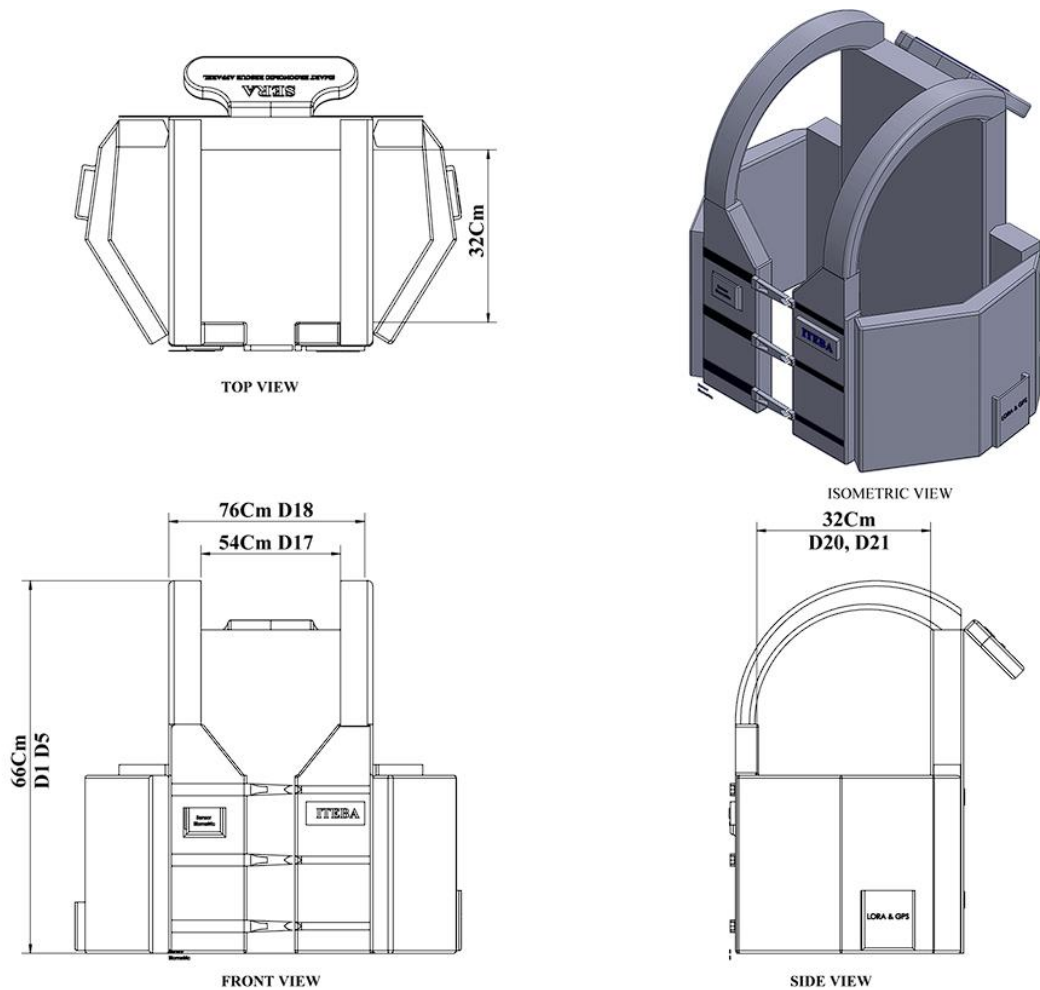


Figure 2. Design of SERA

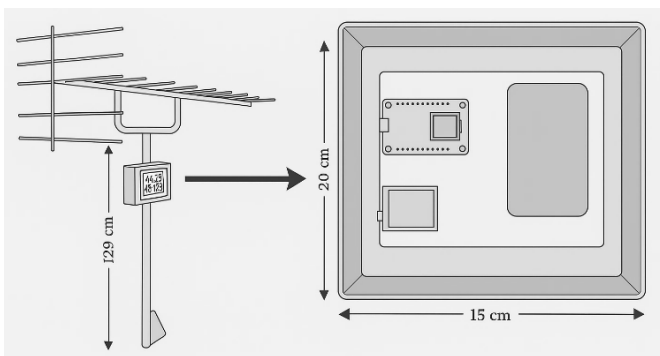


Figure 3. Receiver station design

2.2 Electronical design

Figure 4 illustrates the working process of the LoRa Sender device, which is designed to wirelessly transmit sensor data in a LoRa-based smart buoy system. The ESP32 microcontroller functions as the central control unit, coordinating all modules within the system. It manages communication between components and handles the processing and transmission of data from the connected sensors.

The NEO-M8M GPS module is used to obtain real-time location coordinates. This location data is crucial for tracking the device's position in field applications, especially during remote rescue operations. The coordinate data, along with readings from the DS18B20 temperature sensor and heart rate

sensor, is processed by the ESP32 and then transmitted using the LoRa RFM95 module to a ground-based receiving station. A 3 dBi antenna is employed to enhance the transmission and reception range of the LoRa signal, ensuring stable communication even over long distances.

To ensure autonomous and portable operation in the field, the system is powered by a 5000 mAh power bank as its main energy source. The use of a power bank allows the device to remain active for extended periods without relying on a fixed power supply. With the integration of all these components, the LoRa Sender device is capable of efficiently supporting real-time monitoring and tracking during rescue operations.

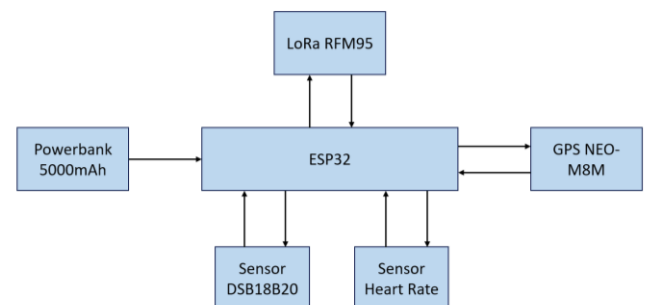


Figure 4. Electronical sender design

The electronic system at the receiving station is designed to receive and manage location data from smart buoy devices.

The ESP32 microcontroller functions as the central control unit of the system. In addition to coordinating communication between modules, the ESP32 is connected to the internet via a Wi-Fi connection, allowing real-time data transmission to servers or monitoring applications.

The system uses an RFM95 LoRa module to receive coordinate data transmitted by the buoy device. To enhance signal stability and range, it is equipped with a 5-element Yagi antenna, which provides higher directional gain than conventional antennas, thereby improving signal reception from distant locations.

As its primary power source, the system utilizes a 5000 mAh power bank. This allows the receiving station to operate independently for extended periods without relying on a fixed power supply. The flow of power and data between components in this system is visually illustrated in Figure 5.

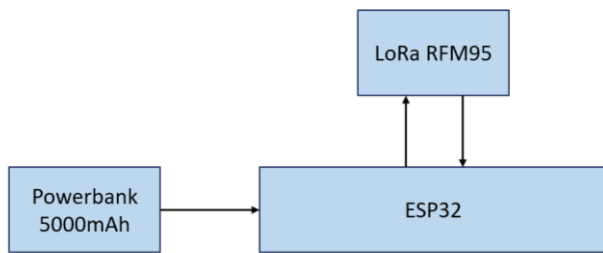


Figure 5. Elektronical receiver design

2.3 Field testing

The development team conducted field testing in July 2025 on Belakang Padang Island, Batam City. The purpose of this activity was to ensure that the smart buoy system could operate optimally under real-world field conditions. During the testing process, the team connected the buoy system to a LoRa receiver station. This station was used to receive and display real-time location data from the buoys, allowing the position of the device to be easily monitored throughout the testing period.

2.4 Data analysis

This study analyzes the SERA system to evaluate the accuracy of the DS18B20 temperature sensor and the heart rate sensor in detecting symptoms of hypothermia. Both sensors function to monitor the user's physical condition in real time. The data obtained is used to identify signs of hypothermia

based on changes in body temperature and heart rate.

In addition to sensor data analysis, this study also considers the Receiver Signal Strength Indicator (RSSI) as an important parameter in wireless communication systems. RSSI is a measure of the strength of the radio signal received by the receiving station. By analyzing the RSSI value, the relative distance between the transmitting and receiving stations can be estimated.

Distance estimation using RSSI is useful for assessing the accuracy of data transmission over a given range between the sender and receiver. RSSI values are expressed in dBm as negative numbers, where values closer to 0 indicate a stronger signal. To calculate the RSSI value, Eq. (1) is used, which is based on a signal propagation model that takes into account distance and surrounding environmental conditions [35].

$$\text{RSSI} = 10n \log_{10} (d) A \quad (1)$$

$$d = 10^{(\text{RSSI}-A)/10n} \quad (2)$$

Eq. (1) is used in this study to calculate the RSSI value based on environmental parameters and communication distance. In this equation, n represents the path loss exponent, d is the communication distance in meters, and A is the reference RSSI value measured at a distance of 1 meter. These three parameters are interrelated in determining the signal strength received by the receiving device.

The equation can then be rearranged into Eq. (2), which is used to estimate the distance between the transmitter and receiver based on the measured RSSI value. By using Eq. (2), the system can estimate distances in a more practical and efficient manner. This equation serves as an important foundation in wireless communication analysis, particularly for determining the position of devices [36].

3. RESULTS

3.1 SERA design

The SERA design integrates lightweight, ergonomic materials that are resistant to extreme environmental conditions. SERA is equipped with smart biometric sensors to monitor body temperature and pulse for early detection of hypothermia, as well as a GPS module for real-time location tracking. Based on the SERA design specifications, the product dimensions are as shown in Figure 6 [37].

The ergonomic dimensions of SERA are determined based on user requirements, as presented in Table 1 [38].

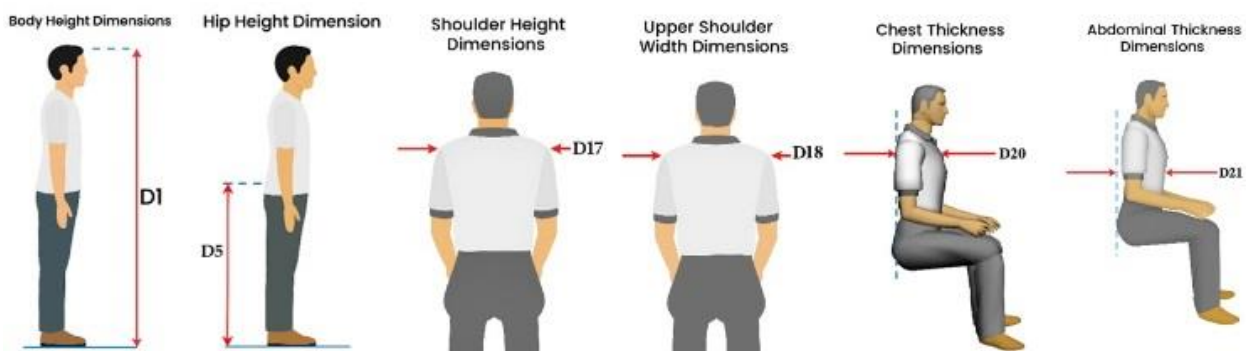


Figure 6. Anthropometric dimensions of the human body

Table 1. Dimensions of SERA

No.	Dimensions	Description
1	Height (D1)	To determine the length of SERA
2	Hip Height (D5)	To determine the length of SERA
3	Shoulder width (D17)	To determine the ideal width of the SERA shoulder
4	Upper shoulder width (D18)	To determine the width of the upper shoulder of SERA
5	Chest thickness (D20)	To determine the ideal chest thickness
6	Thickness of the stomach (D21)	To determine the ideal thickness of the abdomen

The data in this study were analyzed using a precision level of 5% and a confidence level of 95%. Therefore, the data adequacy test was calculated using a value of $k = 2$ and $s = 0.05$. If $N' < N$, the amount of data collected is considered sufficient [39]. Conversely, if $N' > N$, the amount of data collected is considered insufficient. The formula used for the data adequacy test is as follows:

$$N' = \left[\frac{k \sqrt{N(\sum x^2) - (\sum x)^2}}{\sum x} \right]^2 \quad (3)$$

The determination of the upper control limit (UCL) and lower control limit (LCL) was carried out using a 95% confidence level, resulting in a K value of 2.

$$UCL = \bar{X} + k \sigma \quad (4)$$

$$LCL = \bar{X} - k \sigma \quad (5)$$

$$\begin{aligned} BKA &= \bar{X} + k \sigma = 188 + 2 (1.77) = 191.42 \\ BKB &= \bar{X} - k \sigma = 188 - 2 (1.77) = 184.32 \end{aligned}$$

The analysis results indicate that all measurement data D1, D5, D17, D18, D20, and D21 exhibit uniform characteristics and satisfy the adequacy criteria, as presented in Table 2.

Table 2. Measuring data adequacy and uniformity

No.	Measurement	Uniformity	Sufficiency
1	D1	Uniform	Sufficient
2	D5	Uniform	Sufficient
3	D17	Uniform	Sufficient
4	D18	Uniform	Sufficient
5	D20	Uniform	Sufficient
6	D21	Uniform	Sufficient

The anthropometric data were obtained from body dimension measurements of 30 respondents, consisting of traditional fishermen, Search and Rescue (SAR) personnel, and marine medical staff, using the 5th, 50th, and 95th percentiles as shown in Table 3 [40].

Table 3. Respondents and roles

No.	Respondent Groups	Role in Research
1	Traditional fishermen	Primary users, validation of comfort & reliability of tools
2	Search and Rescue Team	Technical evaluation in emergency situations
3	Maritime medical/paramedical personnel	Medical validation of biometric sensors

The calculation was performed using the formula shown in Table 4.

Table 4. Percentile calculation

No.	Percentile	Calculation
1	1%	$\bar{x} - 2.325 \sigma$
2	2.5%	$\bar{x} - 1.96 \sigma$
3	5%	$\bar{x} - 1.645 \sigma$
4	10%	$\bar{x} - 1.28 \sigma$
5	50%	$\bar{x}$
6	90%	$\bar{x} + 1.28 \sigma$
7	95%	$\bar{x} + 1.645 \sigma$
8	97%	$\bar{x} + 1.96 \sigma$
9	99%	$\bar{x} + 2.325 \sigma$

The results of the anthropometric measurements for the respondents are shown in the following Table 5.

Based on the data processing results, all dimensions show a uniform distribution with a relatively small standard deviation, indicating that the data can be considered homogeneous. The minimum and maximum values of all variables are still within the control limits (BKA-BKB), which demonstrates measurement consistency. In addition, the percentile values show a balanced distribution, with P50 close to the mean, while P5 and P95 provide an overview of the range that covers the majority of the population. Therefore, this data can be used as a valid reference in determining product size to match user characteristics.

Figure 7 shows the distribution of values based on percentiles (P5, P50, and P95) of six anthropometric measurement parameters, namely D1, D5, D17, D18, D20, and D21. This visualization illustrates the data distribution, median, and dominant value ranges for each dimension used in the design.

The boxplot results indicate that each dimension has a relatively narrow data distribution, with the median (P50) consistently positioned in the middle of the P5-P95 range. Dimensions D1 and D5 have the largest absolute values because they correspond to height and hip height, while the other dimensions fall within a smaller range according to body characteristics.

Table 5. Results of anthropometric data calculations for respondents

No.	Measurement	Mean	STDV	Min	Max	BKA	BKB	P5	Percentil P50	P95
1	D1	188	1.77	184.94	192.26	191.42	184.33	186	188	191
2	D5	120	2.26	116.78	124.15	124.71	115.65	117	120	125
3	D17	51	1.70	47.62	53.98	54.05	47.23	49	51	54
4	D18	47	1.20	45.24	49.62	49.63	44.83	46	47	50
5	D20	30	0.98	27.92	31.62	31.75	27.82	29	30	32
6	D21	31	1.21	29.17	33.92	33.81	28.98	29	31	32

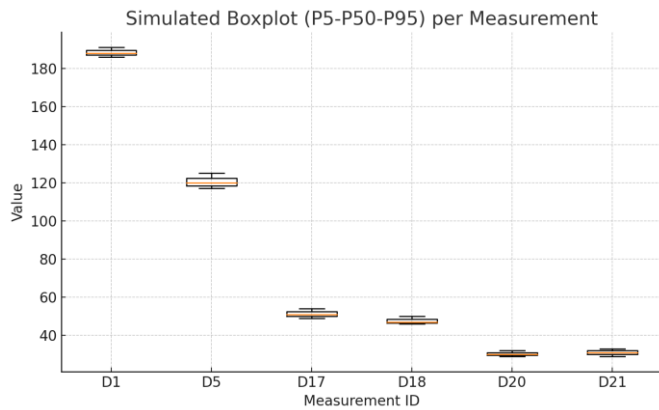


Figure 7. Graph showing value distribution by percentiles

Figure 8 presents a comparison of the mean values with the upper control limit (UCL) and lower control limit (LCL) for six anthropometric measurement parameters.

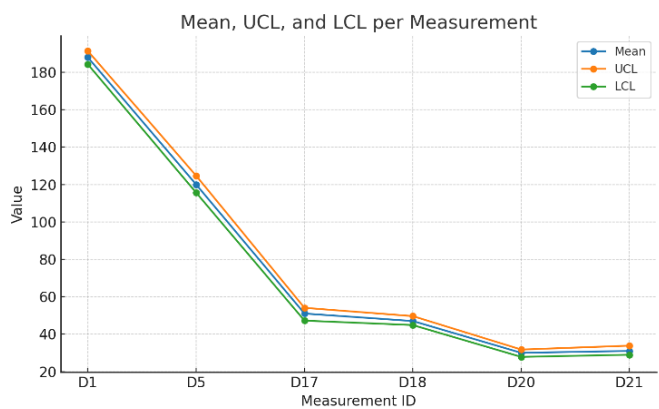


Figure 8. Control chart showing mean, UCL, and LCL

Figure 8 shows that all dimensions have average values within the upper and lower control limits. This indicates that the measurement data is consistent and shows no significant deviations. Dimensions D1 and D5 have the highest values, corresponding to height and hip height, while the other

dimensions are relatively smaller. The closeness between the mean line, UCL, and LCL also indicates low data variation, making the measurement results reliable as a basis for ergonomic design.

In calculating the ideal product dimensions, the 95th percentile is used, representing the average of the measured population with the aim of accommodating all body types, both large and small [41]. The data in Table 6 are used to determine the main size of the life jacket to match the user's body proportions.

Table 6. SERA measurement design based on 95th percentile ergonomics

No.	Measured Data	Description	Size (cm)	Description
1	Body Height	D1	191-	Life jacket length
2	Hip Height	D5	125 =	(body height - hip height)
3	Shoulder Side Width	D17	54	Life jacket shoulder side width
4	Upper Shoulder Width	D18	76	Life jacket upper shoulder width
5	Chest Thickness	D20	32	Chest thickness
6	Abdominal Thickness	D21	32	Abdominal thickness

Based on the data, the length of the life jacket is determined by the difference between body height (191 cm) and hip height (125 cm), resulting in 66 cm. The side shoulder width of 54 cm is used as a reference for determining the shoulder width of the life jacket, while the upper shoulder width of 76 cm enhances the fit of the upper body. Chest thickness (32 cm) and abdominal thickness (32 cm) serve as the basis for designing ergonomic space to ensure that the life jacket does not restrict breathing or movement. Thus, all body dimensions function as important parameters in producing an ergonomic, comfortable, and functional life jacket design. Figure 9 illustrates the results of body dimension measurements relevant to life jacket design.

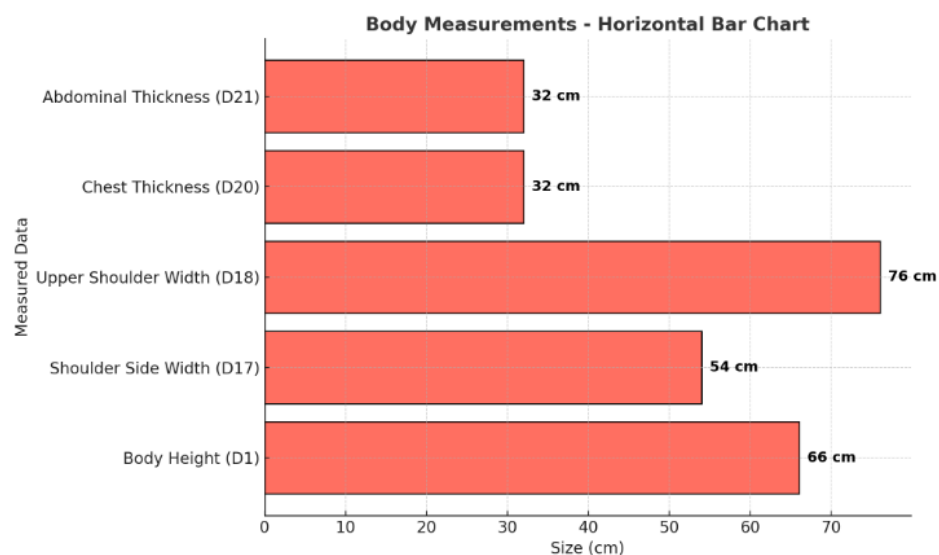


Figure 9. Body measurements

Based on testing, SERA demonstrates excellent and consistent buoyancy performance across a wide range of user weights. The only difference observed is the volume of the jacket submerged in water. For users weighing between 55 kg and 65 kg, approximately 15–20% of the jacket is submerged, while users weighing between 70 kg and 75 kg submerge approximately 25–30%. For heavier users, weighing 77 kg to 90 kg, about 35–40% of the jacket is submerged, and at 95 kg, around 45% is submerged. Despite these differences, all users remain afloat on the water’s surface and do not sink when standing, with their necks and heads consistently above the waterline. The supine position can also be maintained perfectly, demonstrating that the SERA design particularly the headrest and the placement of waterproof electronic devices is effective in ensuring user safety and stability in the water. The testing of SERA is shown in Figure 10.

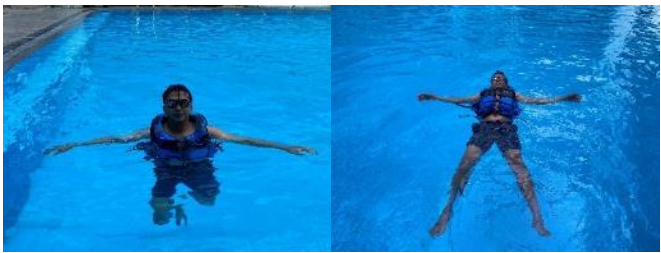


Figure 10. Performance testing of sera

3.2 LoRa sender

The LoRa sender circuit consists of an ESP32 microcontroller serving as the central control unit, connected to a DS18B20 temperature sensor, a heart rate sensor, and a NEO-M8M GPS module. The DS18B20 sensor measures the user's body temperature, while the heart rate sensor monitors the pulse in real time. The NEO-M8M GPS module is responsible for obtaining the user's location coordinates. All

data from these three sensors is collected by the ESP32 and then transmitted wirelessly via the LoRa module to the receiver device. The LoRa sender circuit is shown in Figure 11.

Table 7 shows the connection between the LoRa RFM95 pins and the ESP32.

Table 7. LoRa RFM95 – ESP32

No.	LORA RFM95	ESP32
1	ANA	
2	GND	GND
3	3.3 V	3.3V
4	DIO0	GPIO 2
5	Reset	GPIO 14
6	NSS	GPIO 5
7	SCK	GPIO 18
8	MOSI	GPIO 23
9	MISO	GPIO 19

Table 8 shows the pin connections between the DS18B20 sensor and the ESP32.

Table 8. DS18B20 Sensor – ESP32

No.	DS18B20	ESP32
1	VCC	3.3V
2	Data	GPIO 15 and 10K pull-up resistor
3	GND	GND

Table 9 shows the pin connections between the heart rate sensor and the ESP32.

Table 9. Max30102 Heart Rate Pulse Sensor – ESP32

No.	Sensor Heart Rate	ESP32
1	VCC	3.3V
2	GND	GND
3	A0	GPIO 26

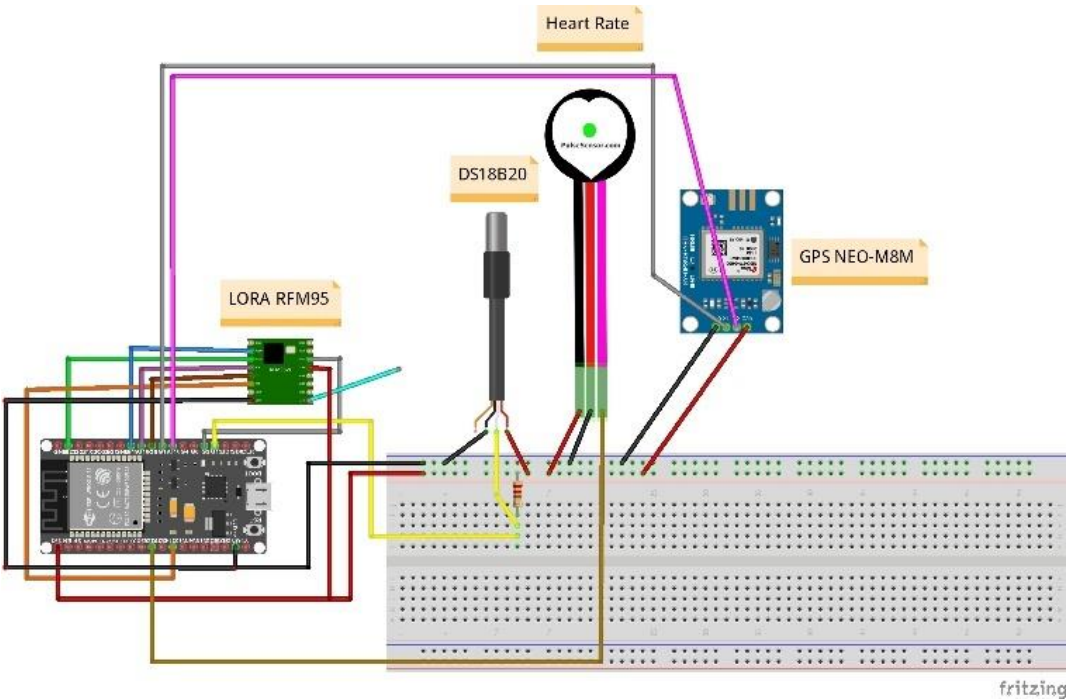


Figure 11. LoRa sender

Table 10 shows the pin connections between the GPS NEO-M8M and the ESP32.

Table 10. GPS NEO- M8M – ESP32

No.	NEO - M8M	ESP32
1	VCC	3.3V
2	GND	GND
3	TX	GPIO16
4	RX	GPIO17

3.3 LoRa receiver

The LoRa receiver utilizes the LoRa RFM95 module and an SD card module as its main components. The LoRa RFM95 module receives measurement data from the LoRa sender, including temperature readings from the DS18B20 sensor and heart rate data from the heart rate sensor. The received data is then automatically stored by the ESP32 in the SD card module, serving as a storage medium to enable further data logging and analysis. The LoRa receiver circuit is shown in Figure 12.

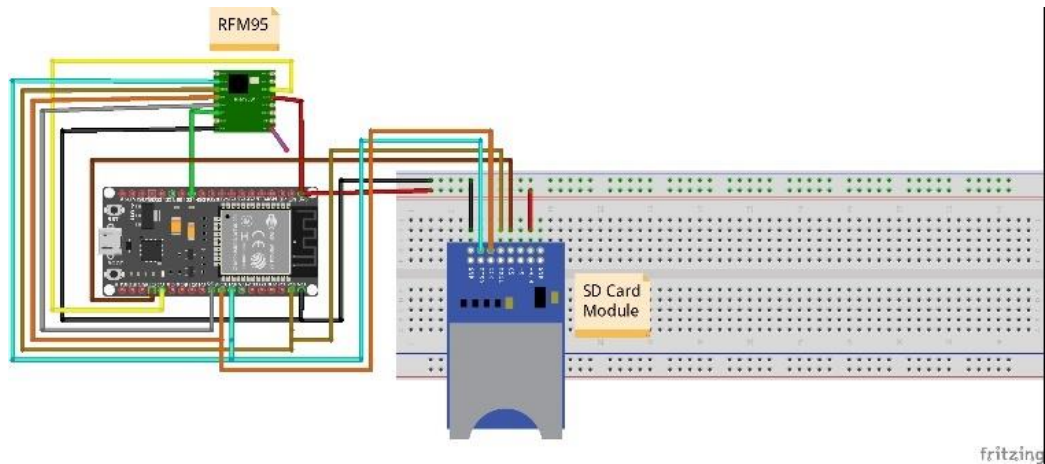


Figure 12. LoRa receiver

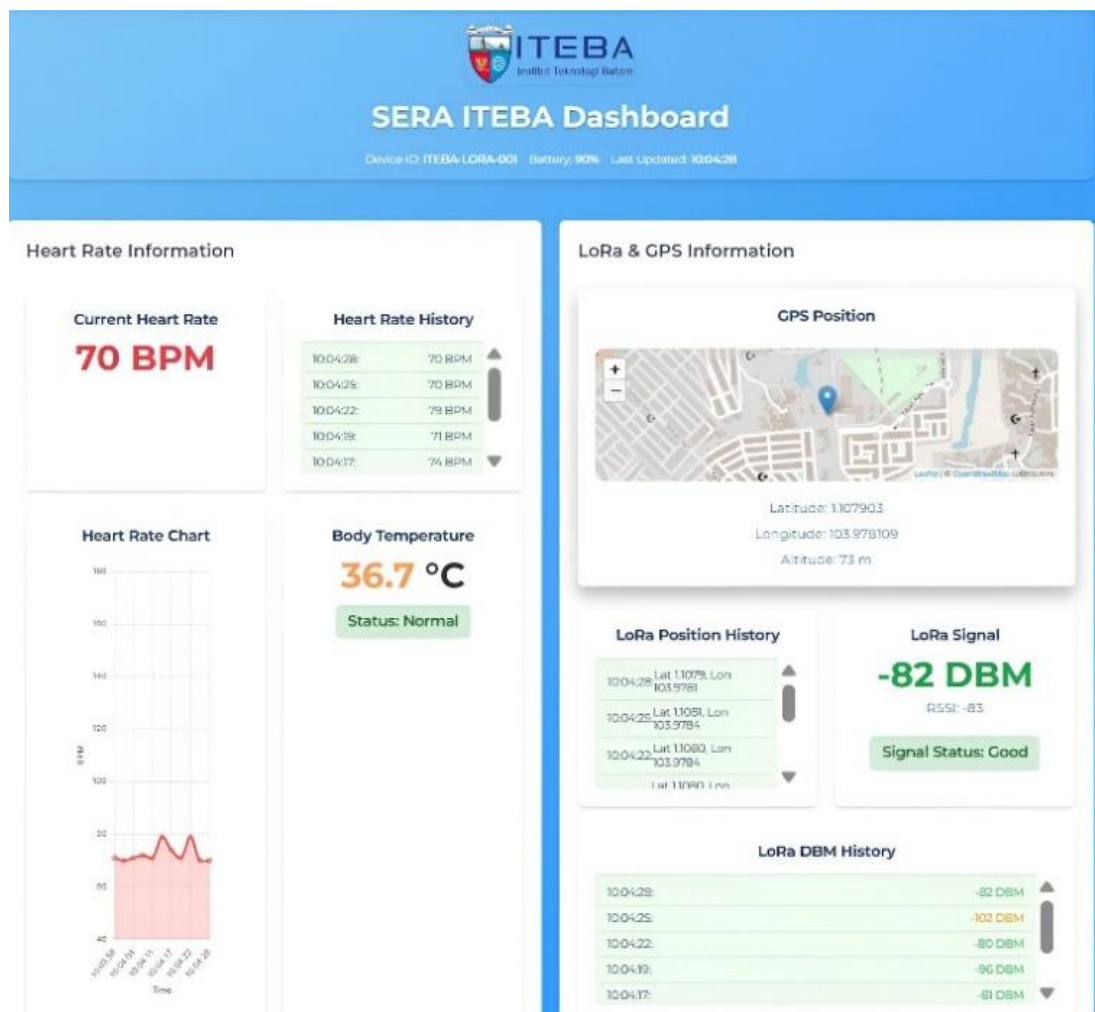


Figure 13. Web view of the SERA system

Table 11 shows the connections between the SD card module and the ESP32.

Table 11. SD card module ESP32

No.	SD Card Module	ESP32
1	3.3 V	3.3V
2	GND	GND
3	CS	GPIO 15
4	MOSI	GPIO 23
5	CLK	GPIO 18
6	MISO	GPIO 19

3.4 Data analysis

The SERA instrument transmits coordinates, RSSI values, body temperature, and heart rate data. The receiving station successfully receives these data and displays them on a webpage, as shown in Figure 13.

Testing of the DS18B20 temperature sensor and the heart rate sensor was conducted to detect symptoms of hypothermia in life jacket users. The DS18B20 sensor monitors body temperature, while the heart rate sensor records heart rate in real time. Both sensors are integrated into an ESP32-based monitoring system installed on the life jacket to detect physiological changes in the body when exposed to cold marine environments. The DS18B20 sensor operates with a sampling frequency of 1 Hz, an error margin of $\pm 0.5^{\circ}\text{C}$ (range -10°C to $+85^{\circ}\text{C}$), and a response time of ~ 750 ms. Meanwhile, the heart rate sensor operates with a sampling frequency of 100 Hz, an error margin of ± 2 BPM, and a response time of less than 1 second.

The testing was conducted over a 90 minutes period using a simulated marine environment scenario designed to induce a gradual decrease in body temperature. During the testing process, body temperature and heart rate data were periodically transmitted via the LoRa module to the receiving station, where the data were stored on an SD card and displayed on a web-based monitoring platform. This system enables continuous tracking of gradual decreases in body

temperature and variations in heart rate, which serve as early indicators of hypothermia symptoms. Table 12 presents the measurement results obtained during the testing of the temperature and heart rate sensors.

Table 12. Testing the DS18B20 and heart rate sensors on SERA

Time (Minute)	DS18B20 Body Temperature ($^{\circ}\text{C}$)	Heart Rate Sensor (BPM)
0	36.8	78
5	36.78	78
10	36.75	79
15	36.73	79
20	36.7	79
25	36.67	78
30	36.62	78
35	36.6	77
40	36.58	77
45	36.52	77
50	36.49	76
55	36.4	76
60	36.38	75
65	36.33	75
70	36.3	75
75	36.29	72
80	36.25	72
85	36.16	72
90	36.05	71

Testing was conducted on life jacket users floating in the sea waters of the Riau Islands at a temperature of 30°C for 90 minutes, with measurements taken every five minutes [42, 43]. The DS18B20 sensor recorded body temperature, represented by a solid blue line, while the heart rate sensor recorded heart rate, represented by a solid red line. The dashed blue line indicates the threshold for mild hypothermia in the temperature range of 36°C - 36.4°C [44], while the shaded blue area represents the period when body temperature fell below 36.4°C , indicating mild hypothermia. The results showed a decrease in body temperature to 36°C , confirming the occurrence of mild hypothermia, as shown in Figure 14.

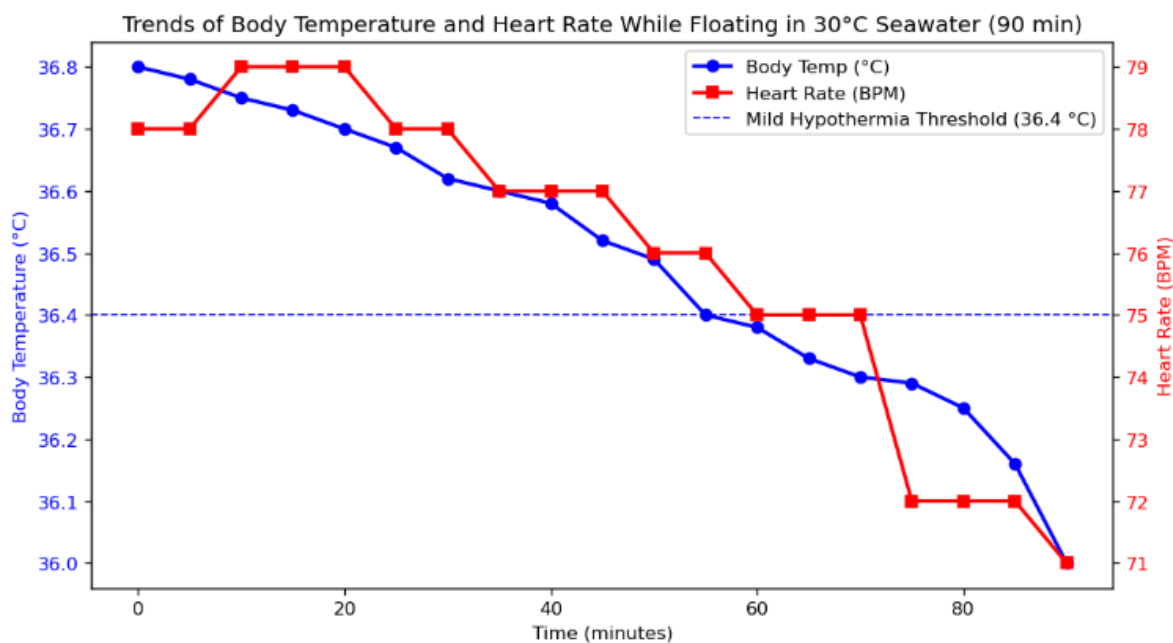


Figure 14. Measurement results of the temperature sensor and heart rate sensor

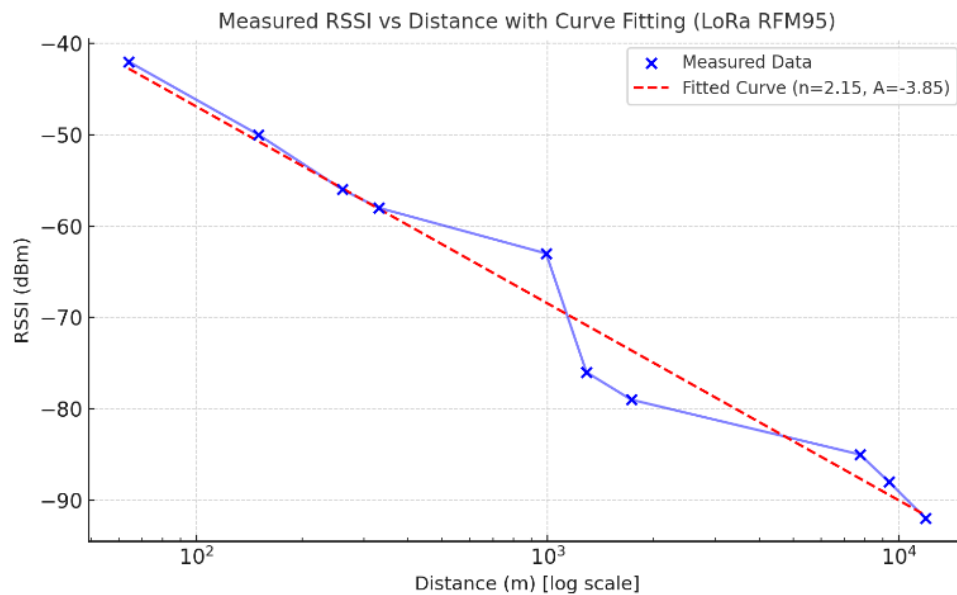


Figure 15. Relation between RSSI value and distance

Table 13. Testing the DS18B20 and heart rate sensors on SERA

Starting Point	Destination Coordinates	Distance (m)	RSSI (dbm)
	1.1556519887948804, 103.89403380755512	64	-42
	1.1556831382410362, 103.89478944139891	150	-50
	1.1570416745742707, 103.89540186419498	260	-56
	1.157312743958163, 103.89605256341584	330	-58
1.15579524 726097, 103.893467 68785762	1.1647694956557653, 103.8928344556692	990	-63
	1.1669684517564318, 103.89009860251855	1290	-76
	1.17055868170281, 103.8883045371556	1730	-79
	1.1701111898437184, 103.82528157160937	7750	-85
	1.1458317743658784, 103.80976174841416	9380	-88
	1.1426567200897284, 103.78695453789548	11940	-92

RSSI testing was conducted in open waters without obstacles using a tracking method, where coordinates and RSSI data were transmitted every minute. The researchers used the LoRa 433 MHz frequency, which is standard in the Asian region. This frequency offers advantages such as a longer signal range, better penetration through obstacles, and greater suitability for large areas such as open waters [45]. The receiving station was located at the Belakang Padang Domestic Port.

Distance determination was performed after the coordinate data was successfully received from SERA. The researchers used the website <https://boulter.com/gps/distance/> to calculate distances by entering the coordinates of the receiving station and the instrument location. Measurement results are presented in metres for distance and in dBm for RSSI values. RSSI values closer to zero indicate a stronger signal, while

values further from zero indicate poorer signal quality.

Figure 15 shows the relationship between distance and RSSI values obtained from the LoRa RFM95 test. The blue dots represent the measurement data, while the dashed red line represents the fitted curve using the log-distance path loss model. The RSSI value decreases as the distance increases, from -42 dBm at 64 meters to -92 dBm at 11,940 meters. The fitting results yield a path loss exponent of 2.15 and a reference RSSI of -3.85 dBm, indicating propagation conditions close to free space. Figure 15 demonstrates that the LoRa RFM95 can function effectively at distances of up to 11,940 meters.

During testing, the highest signal strength (RSSI) was recorded at -42 dBm at a distance of 64 metres, while the lowest signal strength was -92 dBm at a distance of 11,940 metres, which was also the farthest point reached during the test. All data, including RSSI values, distances, and coordinates of the safety buoys, are presented in Table 13.

4. CONCLUSION

The SERA design used anthropometric data from 30 respondents to tailor its size to the user. Its key features include biometric sensors to monitor heart rate and body temperature, a visual early warning system using LED lights, and emergency signal transmission via LoRa technology to detect symptoms of hypothermia. In this study, the researchers equipped the LoRa-based SERA with a GPS module to obtain coordinate positions and a LoRa module to transmit collected data packets. The results showed a relationship between distance and RSSI values, where RSSI values tended to decrease or move away from 0 as the distance between the receiving station and the SERA increased. The effectiveness of this design was demonstrated by the maximum distance achieved by the LoRa module, which reached 11,940 meters (11.94 km). Furthermore, the temperature and heart rate sensors installed on the SERA proved capable of detecting symptoms of mild hypothermia, as evidenced by graphs showing a decrease in body temperature and heart rate during a 90-minute sea trial. However, this system has drawbacks, including a battery life limited to 12 hours and interface

limitations, as there is no visual indicator in the form of a heat map on the website to display the user's condition more intuitively.

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